Experimental and Numerical Investigations on Metal-Inert Gas Wire Arc Additive Manufacturing (Multi-Material & Hybrid Strategies)

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- **WAAM** stands for Wire Arc Additve Manufacturing
- Controlled Welding and layer-by-layer deposition
- Can employ wide range of materials. Printing depends of type of system used, filler wires, nozzle, wire properites, etc.
- Near-Net Shaped and Support Free AM
- Theorized in 1920 with patent granted to Baker *[METHOD of MAKING DECORATIVE ARTICLES]* in 1925.
- Extensive R&D by Cranfield University since 2000s.

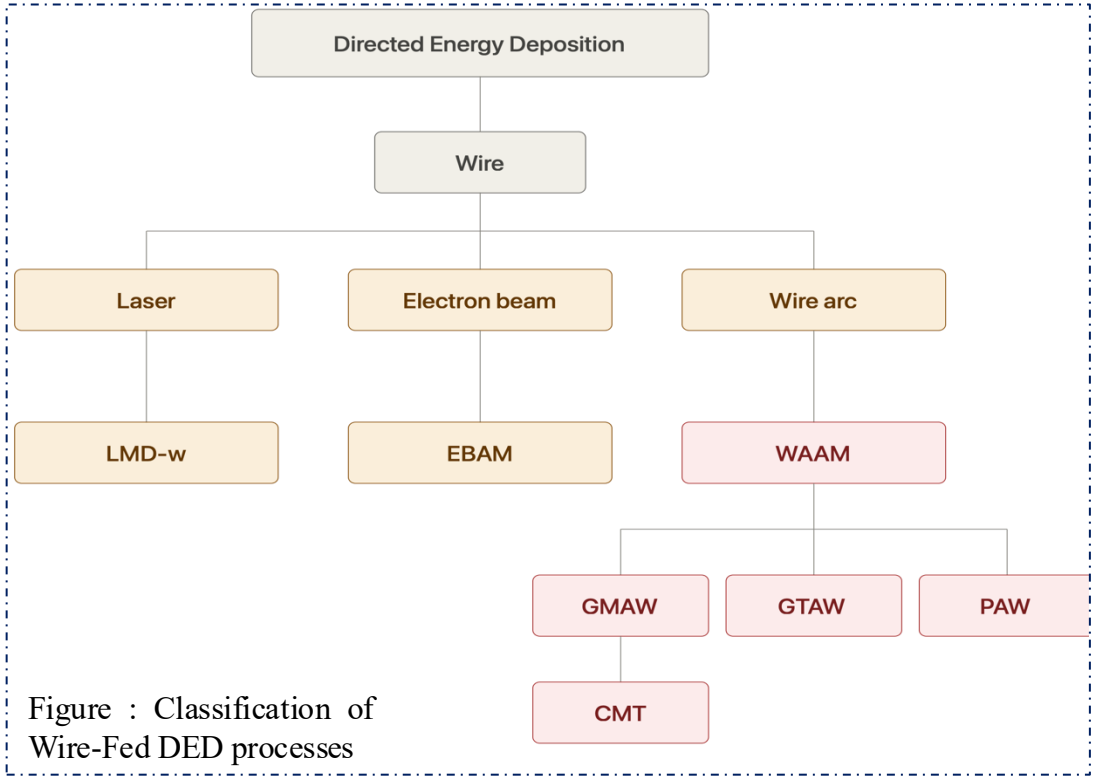


Figure : Classification of Wire-Fed DED processes

Table : Characteritics of several WAAM process adapted from [Guo et al. 2025]

GMAW-based	GTAW-based	PAW-based	CMT-based
- Consumable Electrode Fed into Molten Pool - Suitable for medium to large structures requiring faster build rates - **Deposition : 3-4 kg/h**	- Non-consumable tungsten electrode + Separate wire feed system - Suitable for thin-walled structures and fine-feature parts - **Deposition : 1-2 kg/h**	- Utilizes a non-consumable tungsten electrode with a constricted plasma arc and separate wire feed. - Suitable for high-precision deposition with deeper penetration and finer beads. - **Deposition : 2-4 kg/h**	- Advanced GMAW process with reciprocating wire motion for controlled droplet transfer. - Suited for low-distortion, thin-walled components and dissimilar material joining - **Deposition : 2-3 kg/h**

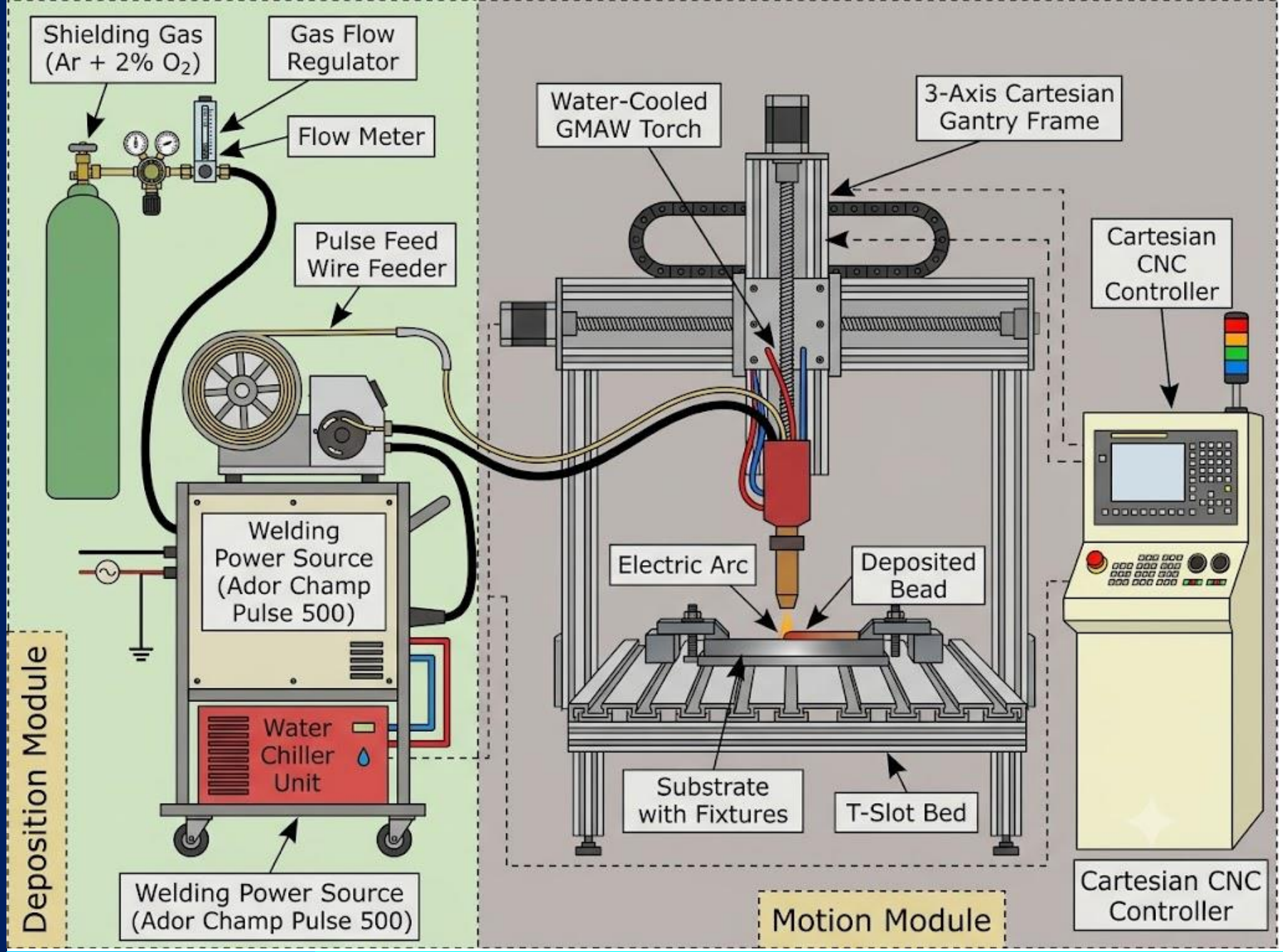
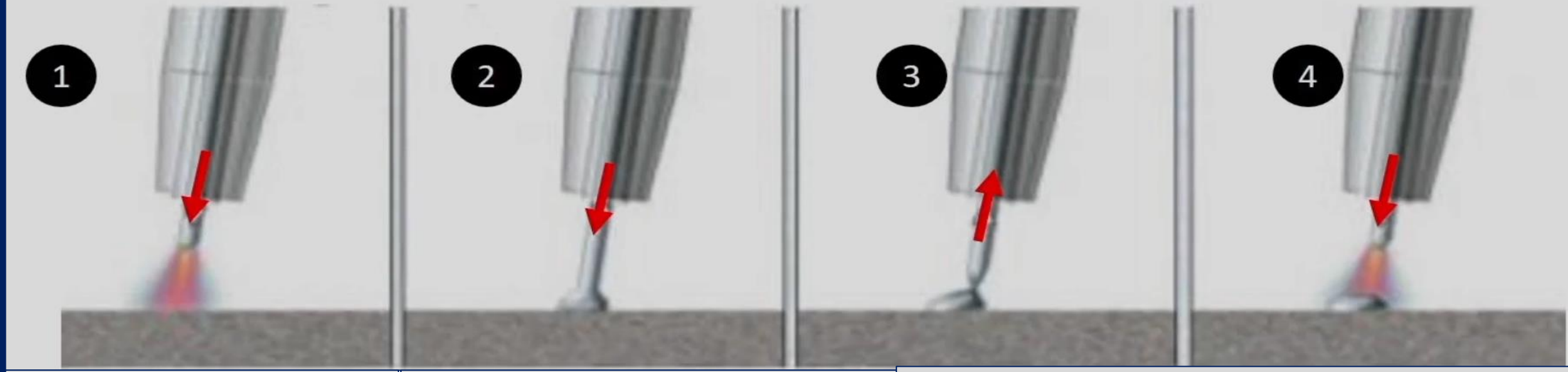
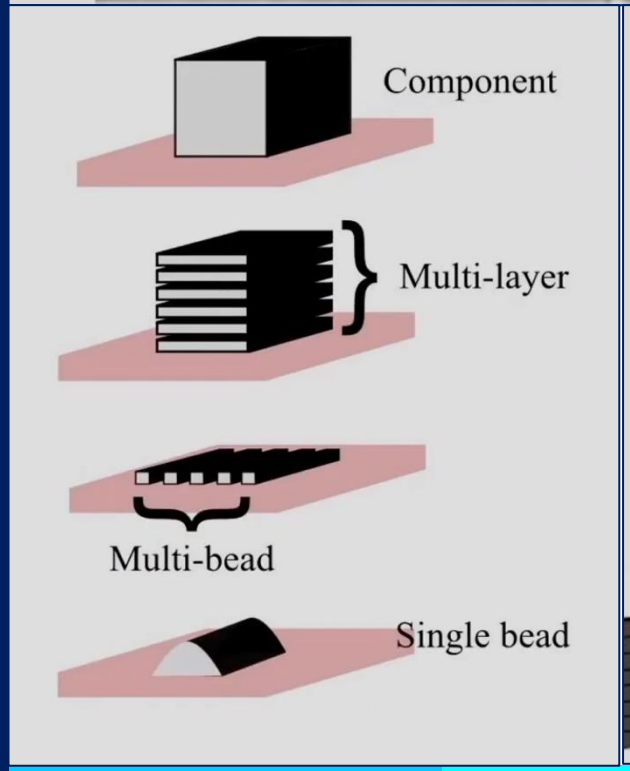
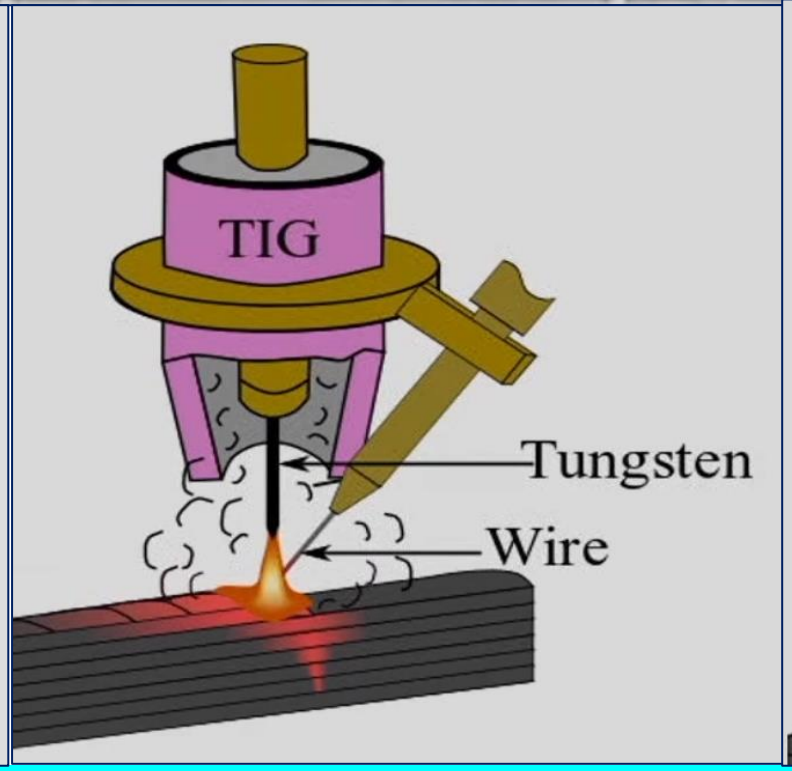
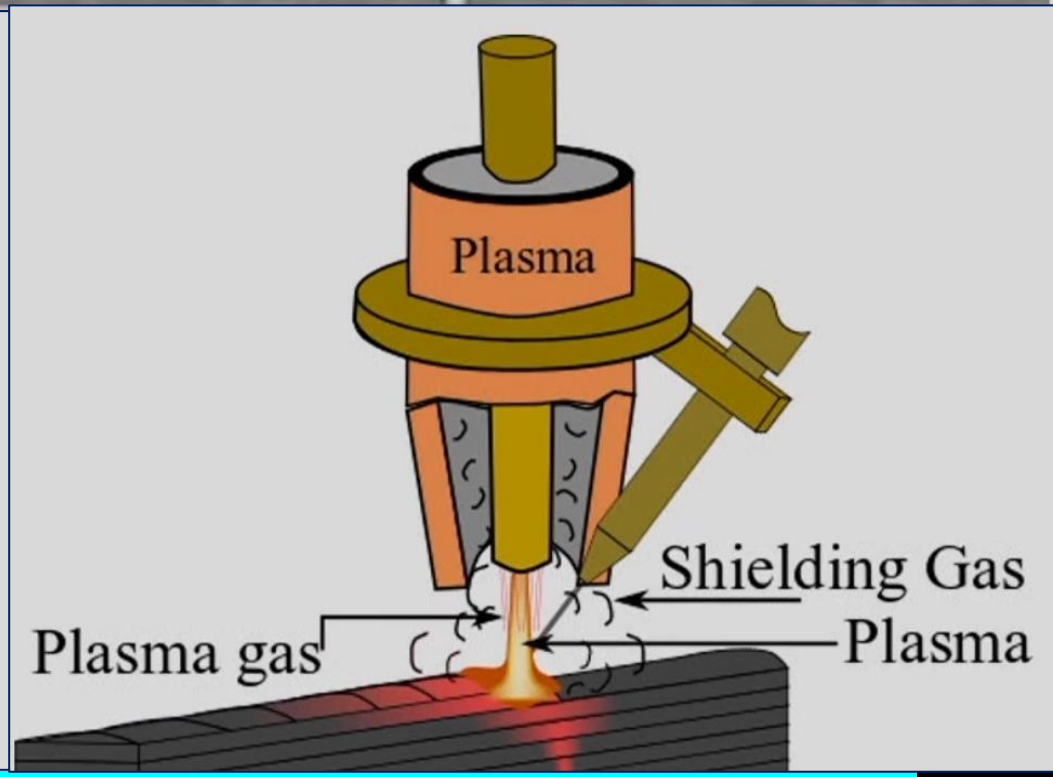


Fig : Schematic of WAAM system at IIST


1
2
3
4


Component
Multi-layer
Multi-bead
Single bead


TIG
Tungsten
Wire


Plasma
Shielding Gas
Plasma gas
Plasma

SS308L

- 308 stainless steel is an austenitic stainless steel, a low carbon nickel-chromium stainless steel with excellent corrosion resistance and good welding performance.

- The low carbon content (C≤0.08) of 308 stainless steel ensures its good resistance to intergranular corrosion, which gives it excellent performance in high-temperature environments.

- 308 electrode is mainly used for welding some low-carbon steel and low alloy steel, such as ordinary carbon structural steel, and high-quality carbon structural steel. It can also be used to weld some stainless steels, including austenitic stainless steels such as 304 and 316. 308 electrode is a stainless steel electrode for welding austenitic stainless steel of 308 series.

Property	Value
Density	7.93 g/cm³
Melting point	1375-1450°C
Thermal expansion	17 x 10^{-6} per K in the range 20-100°C
Thermal Condcutivity	16.3 W/m.K at 100
Poisson's ratio	0.27-0.30
Specific heat	500 J/kg.K
Resistivity	7.1 x 10^{-7} Ω-m

Property	Value
Tensile strength	593 MPa
Yield strength	207 MPa
Elastic modulus	190-210 GPa
Poisson's ratio	0.27-0.30
Elongation at break (in 50 mm)	48%

Element	Content (%)
Iron, Fe	Balance
Chromium, Cr	19-21
Nickel, Ni	10-12
Manganese, Mn	2
Silicon, Si	1
Phosphorous, P	0.045
Carbon, C	0.03
Sulfur, S	0.03

Source : https://www.azom.com/ **&** https://www.dsstainlesssteel.com/

• Multi-material WAAM offers massive deposition rate advantages (2–10 kg/h) and cost efficiency for large multi-metal parts compared to powder AM. Requires Reliable WAAM systems. Preferrable is CMT based

Direct joining of highly dissimilar pairs (e.g. Ti-Al, Fe-Al, Al-Si) forms continuous brittle intermetallic layers ($FeAl_2Si$, $FeAl_3$, Fe_3Al, $TiAl_3$, $NiTi_2$) that crack spontaneously.

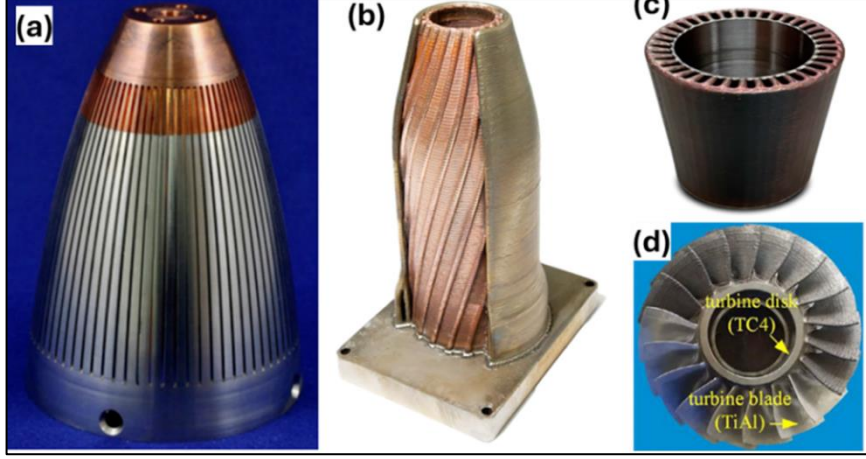


Figure : Use of Bimetallic Materials

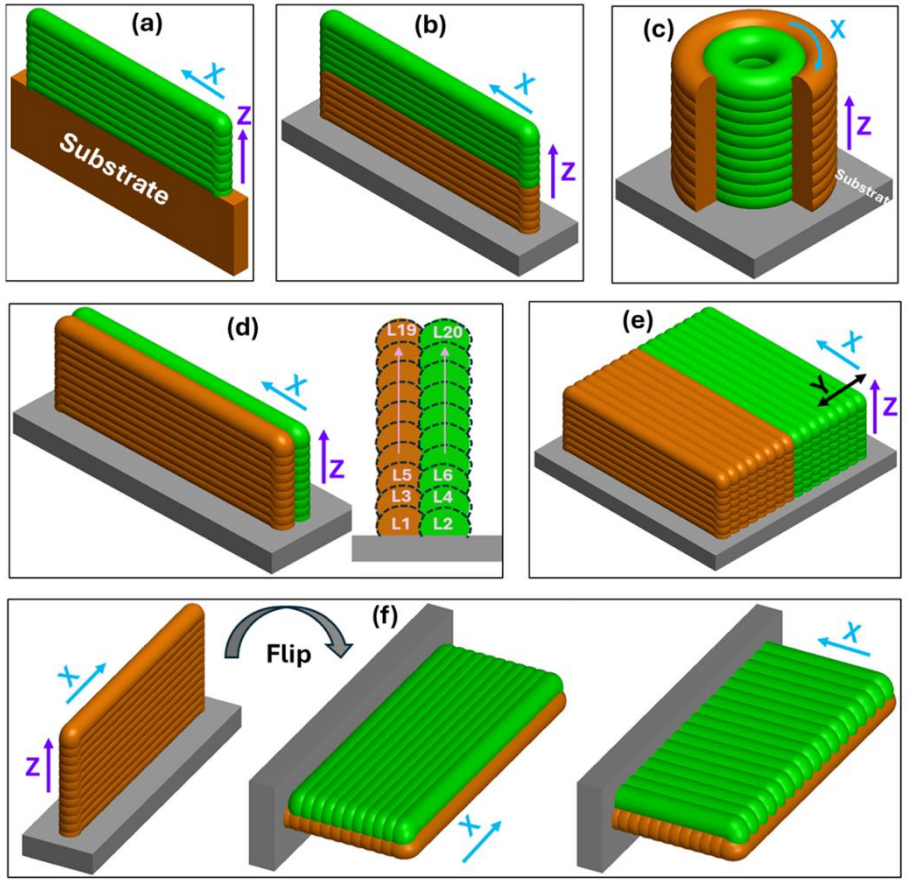
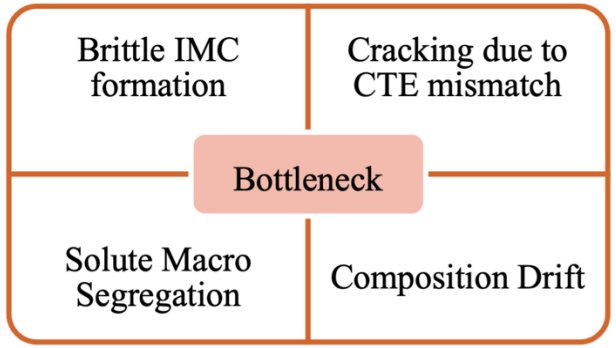


Figure : Deposition Strategies in Bimetallic Structure Fabrication using AM

Thermal expansion mismatch induces severe interfacial shear residual stresses during cooling.

• Designing intermediate functionally graded transition layers or compatible buffer metals (e.g. Ti-V-Cr-Fe or Cu-Ni-Steel) eliminates brittle IMCs.

• Active thermal control (low heat input CMT mode, interpass cooling, ultrasonic assistance) suppresses excessive liquid-state interdiffusion



- Interfacial reactions depend on material-pair chemistry, dilution, peak temperature, and time at elevated temperature.
- Brittle intermetallic compounds (IMCs) can reduce local ductility and promote crack initiation; their presence alone does not establish spontaneous fracture.
- Thermal expansion mismatch and solidification shrinkage contribute to residual stress, cracking, and delamination.
- Elemental migration can create local hardness gradients and alter the mechanical response across the interface.
- **MATERIAL DESIGN, THERMAL MANAGEMENT, PROCESS CONTROL & PREDICTIVE SCREENING** needs to be employed

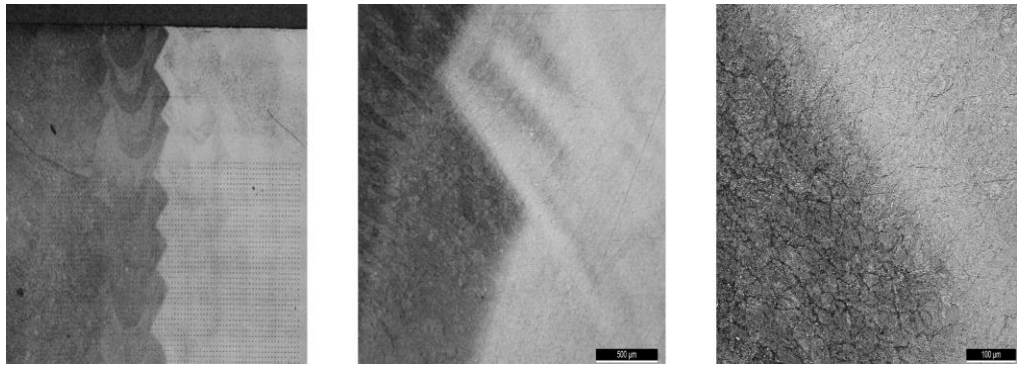


Figure : SEM Images of the H-13 to 410 NiMo Interface

EVIDENCE FROM EXPERIMENTAL WAAM STUDIES	
Material System	**Reported Result**
Ti6Al4V / Al6.21Cu, with 0.3 mm Nb foil	UTS increased from 94.5 to 120.9 MPa; elongation increased from 0.34% to 1.14%
ER50-6 steel / HS211 silicon bronze, Cu–Ni interlayer	UTS increased from 207.5 to 345.2 MPa
Steel / aluminium	$FeAl_2Si$ intermetallic regions observed using SEM and EBSD
6060 / 5087 aluminium	Transition zone approximately 15–65 µm; transition zone did not limit component strength in the reported study

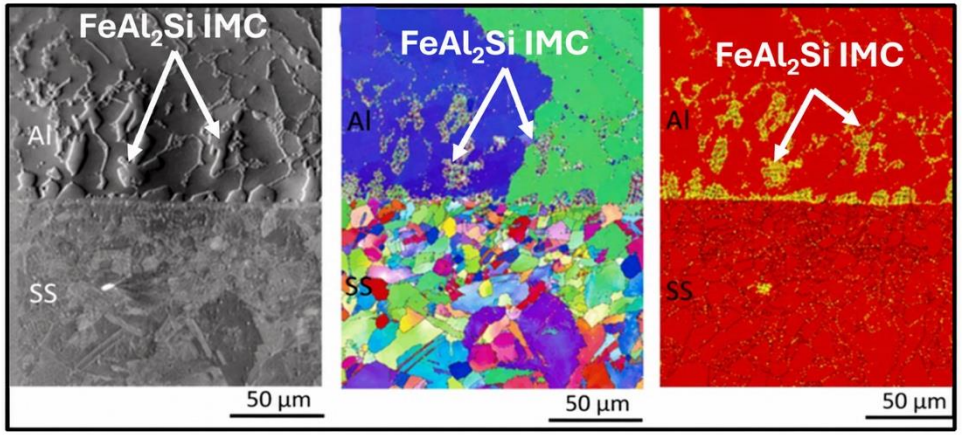


Figure : SEM images., EBSD phase maps for Steel/Aluminum BS showing FeAl2Si IMC

ULTRASONIC

- Ultrasonic vibration assisted welding involves introducing high frequency mechanical vibrations into the workpiece, welding arc, filler wire, or molten weld pool.
- UVA welding modifies these conditions through acoustic cavitation, acoustic streaming, oscillatory pressure fields, and mechanical interaction with the solidifying microstructure .
- The resulting effects depend on the ultrasonic coupling configuration, frequency, amplitu de, acoustic power, material properties, and location of vibration application.

Solidification is governed thermal grad ient solidification rate, melt pool conve ction, alloy composition, and repeated nucleation and growth of crystals.

Several ways to deliver the vibration exist:
a transducer under the substrate,
a probe dipped into the pool,
a pulsed arc at ultrasonic frequency,
a roller on the plate,
or an impact gun after the arc

A weld pool is small, shallow and strongly superheated. The sound travels only a short distance through liquid, and the pool sits close to the solidification front where dendrites form.

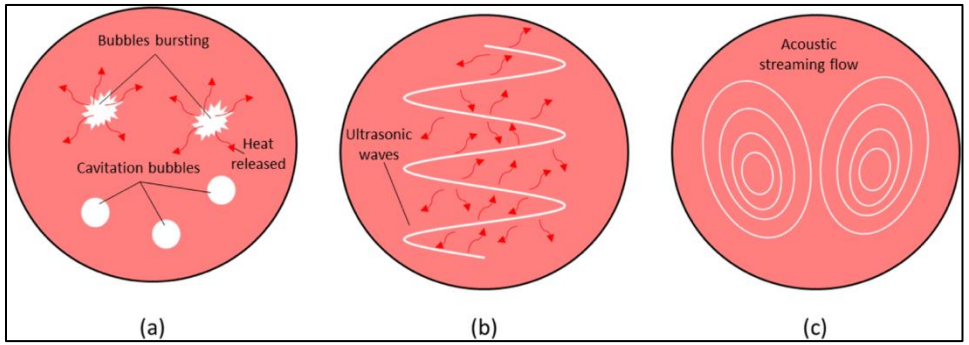
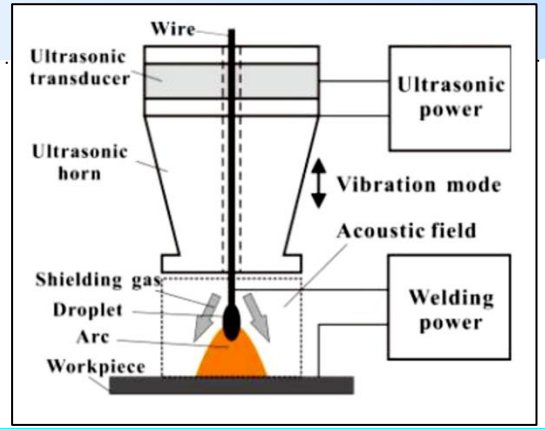


Figure : Ultrasound Induced Phenomena

ABOUT DEVICE

- Ultrasonic generator is a solid electronic box which is fed in with 220 V and 50 Hz from the mains as inputs and transforms the input into 800–1000 V and 20 kHz electrical supply.
- Ultrasonic transducer or converter is a devices which converts ultrasonic waves to the desired electrical signals or vice versa.
- Owing to the negligible amplitude at the end face of transducer, ~0.001–0.1 mm, the horn is integrated into the ultrasonic setup so as to enhance the amplitude of ultrasonic vibrations.
- The main function of booster is to amplify the resultant vibration amplitude of transducer.



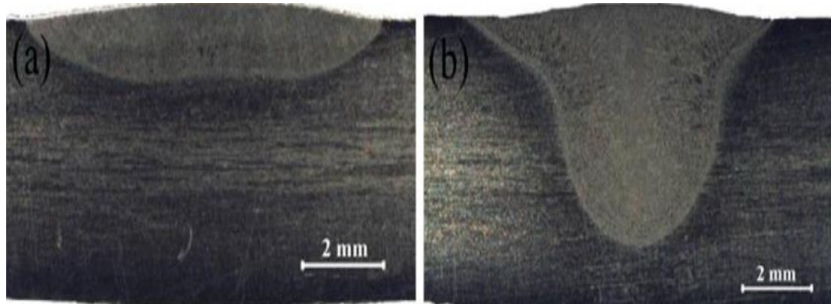


Figure : Weld penetration of (a) conventional arc welding and (b) ultrasonic wave-assisted arc welding

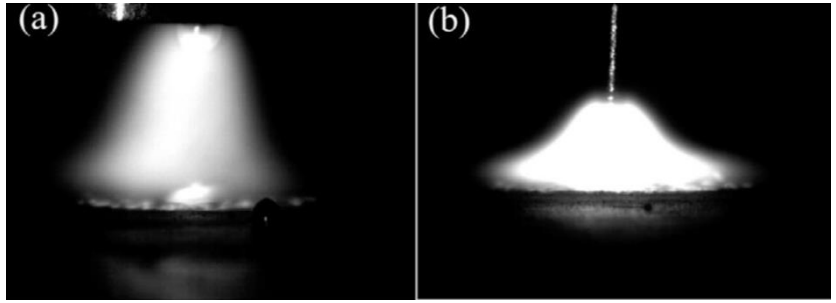


Figure : Arc shape of (a) conventional arc and (b) ultrasonic wave-assisted arc

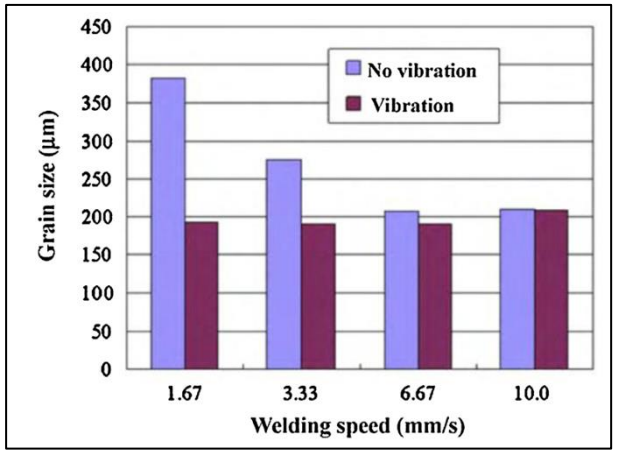


Figure : Average grain size of the weld metal made without and with ultrasonic vibration using various welding speeds

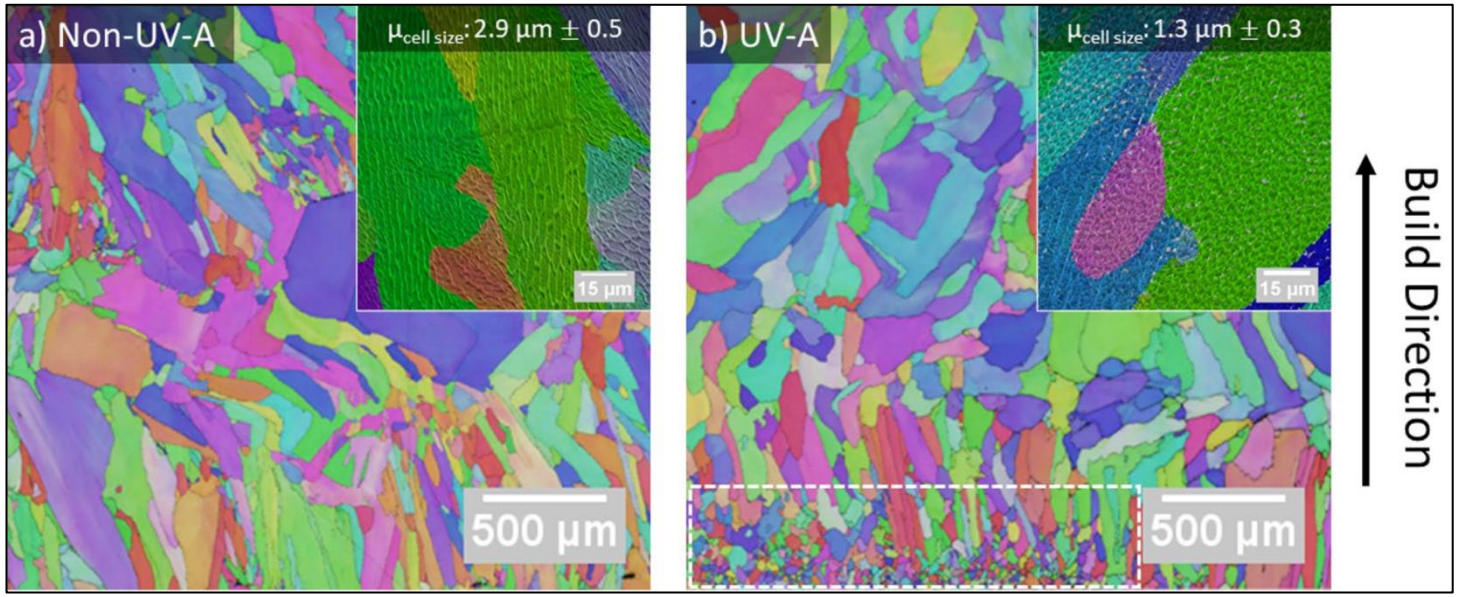


Figure : Representative EBSD inverse pole figures of the as-deposited 316L samples near the substrate

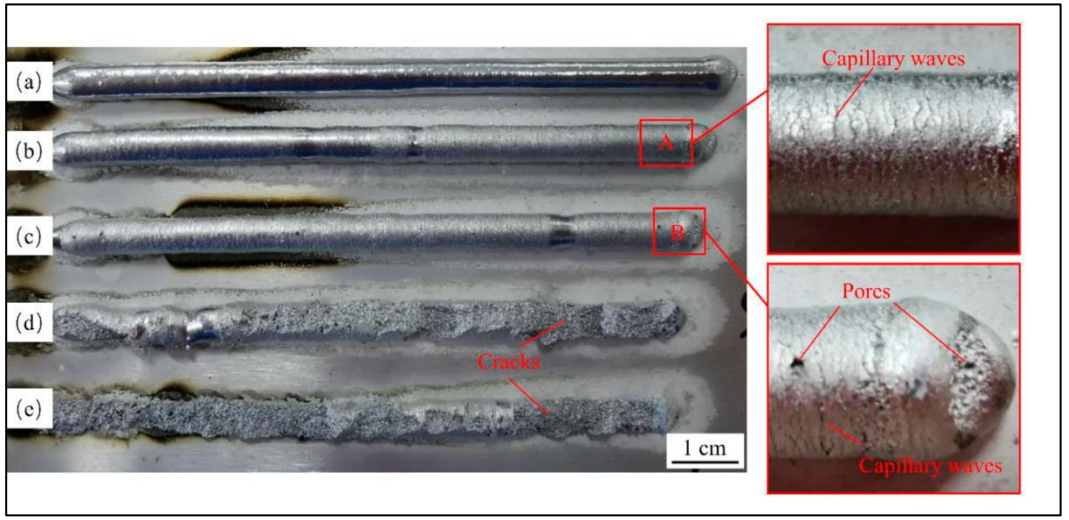
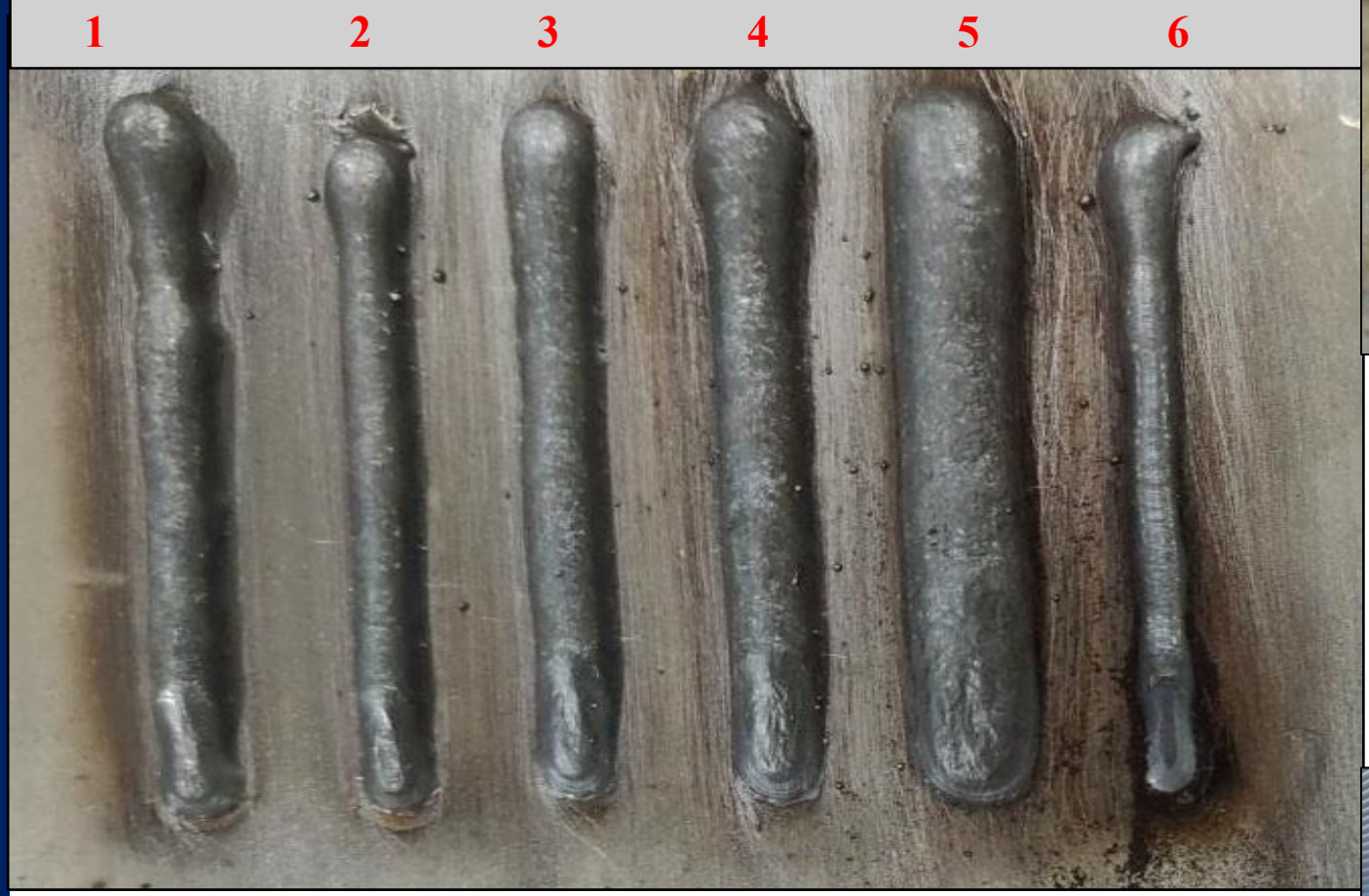


Figure : Effect of different ultrasonic amplitudes on weld:

BEAD TRIALS





Material : SS308L || Diameter : 0.8 mm

Abruptly stopped in between during printing of multilayer walls. Observed wire pulling out of the driving roller. Key suspect was diameter and it getting stuck with liner.



S/N	Current	Voltage	Wire Feed	Torch Speed	Bead Height	Bead Width
1	120 A	19 V	9.5 m/min	300 mm/min	3.78 mm	5.5 mm
2	120 A	19 V	9.5 m/min	400 mm/min	3 mm	4.72 mm
3	140 A	19.7 V	11.9 m/min	400 mm/min	3.25 mm	5.64 mm
4	140 A	19.7 V	11.9 m/min	300 mm/min	3.78 mm	6.42 mm
5	200 A	24.2 V	19.9 m/min	300 mm/min	4.11 mm	9.4 mm
6	70 A	25.2 V	-	-	2.5 mm	4.08 mm

BEAD TRIALS







S/N	Current	Voltage	Wire Feed	Torch Speed	Bead Height	Bead Width
1	120 A	19.7 V	6 m/min	300 mm/min	-	
2	120 A	19.7 V	6 m/min	500 mm/min	2.6 mm	7 mm
3	140 A	19.7 V	6 m/min	600 mm/min	-	
4	140 A	19.3 V	4.8 m/min	300 mm/min		
5	200 A	19.3 V	4.8 m/min	500 mm/min		
6	70 A	19.3 V	5.3 m/min	300 mm/min		

Material : SS308L || Diameter : 1.2 mm

- Abruptly stopped in between during printing of multilayer walls.
- Wire stayed in place, but process faced severe fluctuation of current – voltage in between, once again diverting attention towards wire and its compatibility with nozzle.

- FEM simulations helped in analysing the deposition strategy for optimal heat build up and residual stress. Spiral Pattern promotes heat accumulation. Von Misses Stress is found to be higher towards end irrespective of patterns, and lower for high scan speeds.

- Continuous Deposition led to Slower cooling rate, coarser columnar grains, large SDAS, and reduced Hardness. Dwell Time allows smaller SDAS. Cooling Rates can be obtained from FEM simulation which can be used to determine Secondary Dendrite Arm Spacing [For SS308L SDAS = $50CR^{-0.4}$]

- CFD needs to be employed to capture Melt pool, Marangoni forces, and other flow related physics like acoustic streaming, cavitation, etc.

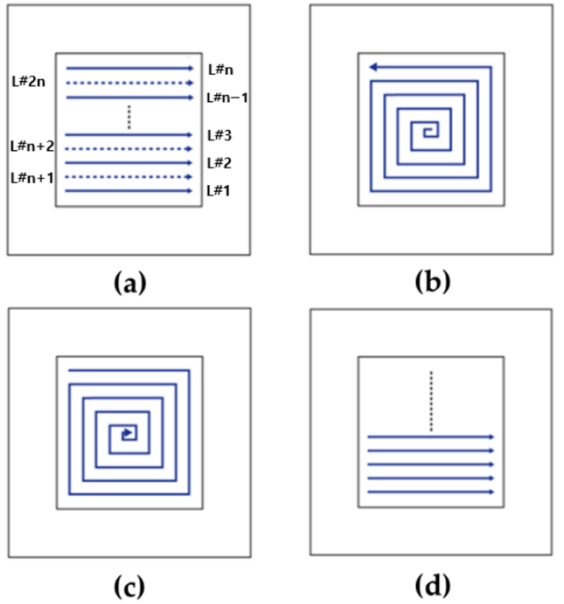


Figure : Line and Spiral Strategy

FEM	FDM
- FEM with element birth-and-death can't predict wall thickness or sagging - Allows non-uniform meshes - Can incorporate multiple BC types, variety of softwares.	- Lower Computational Cost - Morphology Resolution i.e sagging, overlapping - Quick Results - Limited to Structured Cartesian Grid

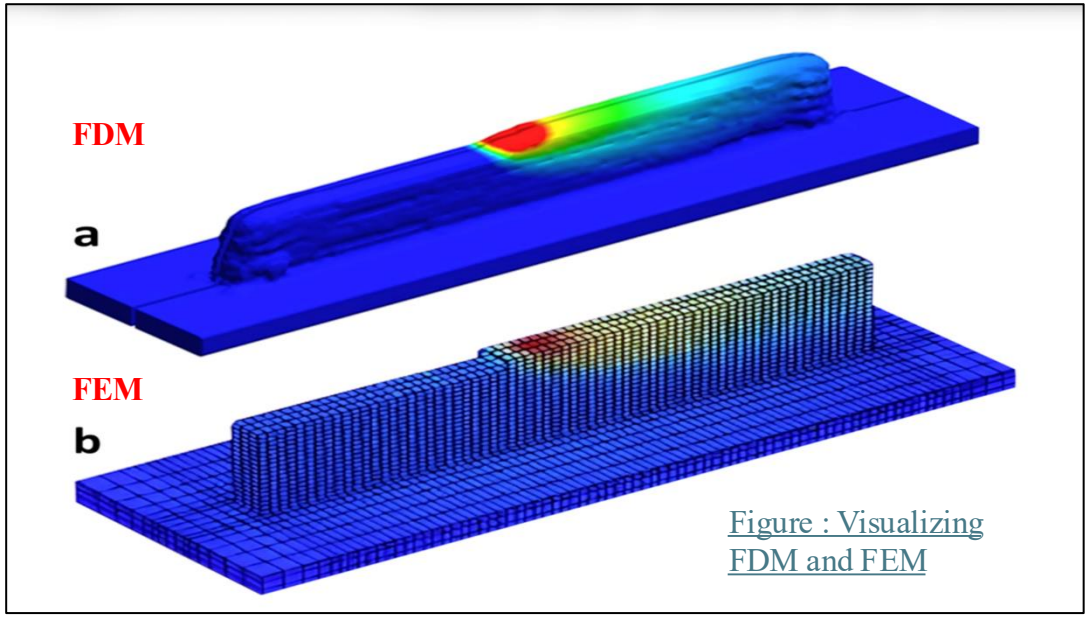


Figure : Visualizing FDM and FEM

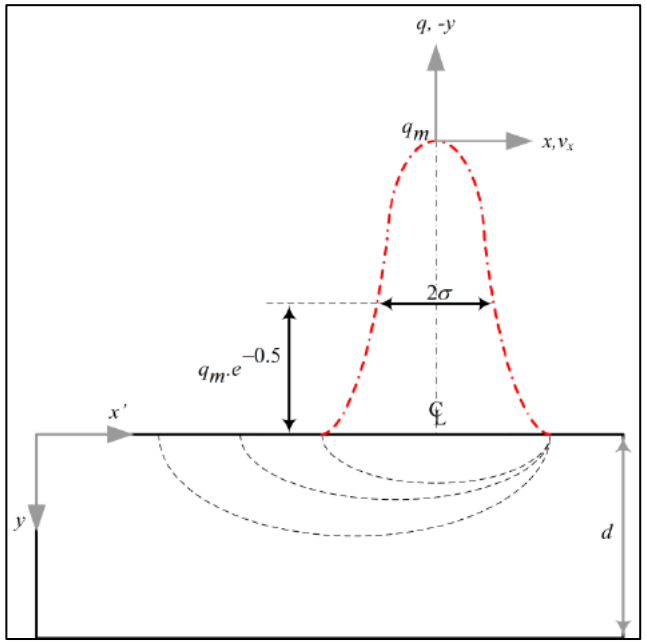


Figure : Gaussian Heat Source

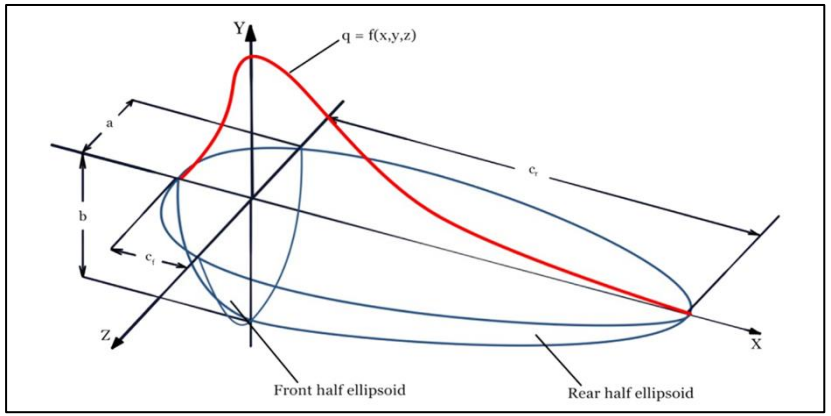


Figure : Goldak Double Ellipse

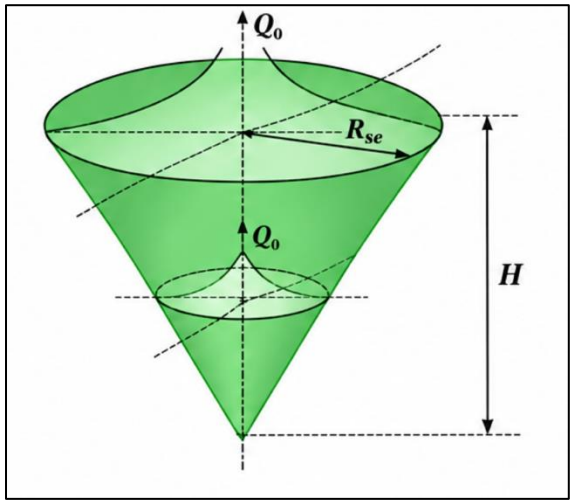


Figure : Conical Heat Source Model with Depth Dependent Radius

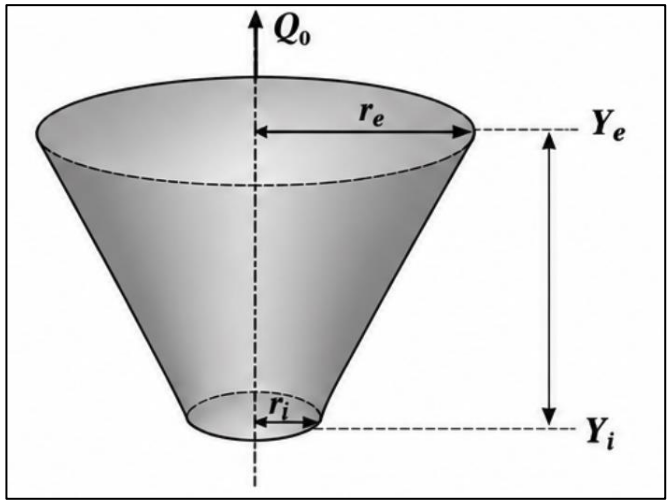


Figure : Conical Heat Source

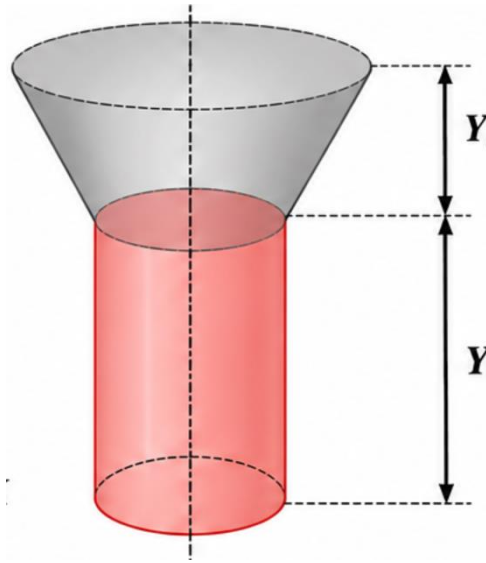


Figure : Cylindrical Keyhole

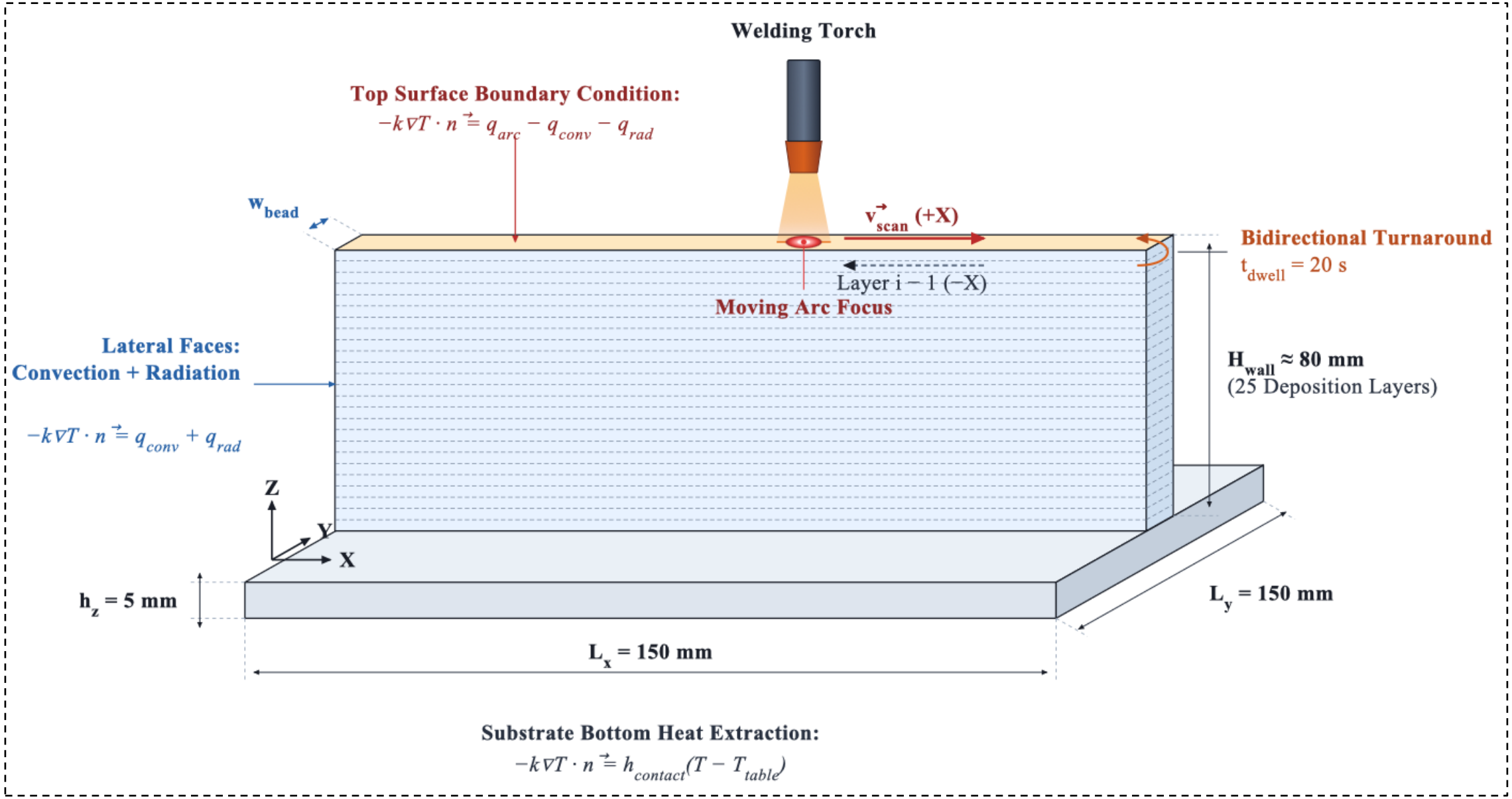


Figure : Simulation Domain and Governing Equations

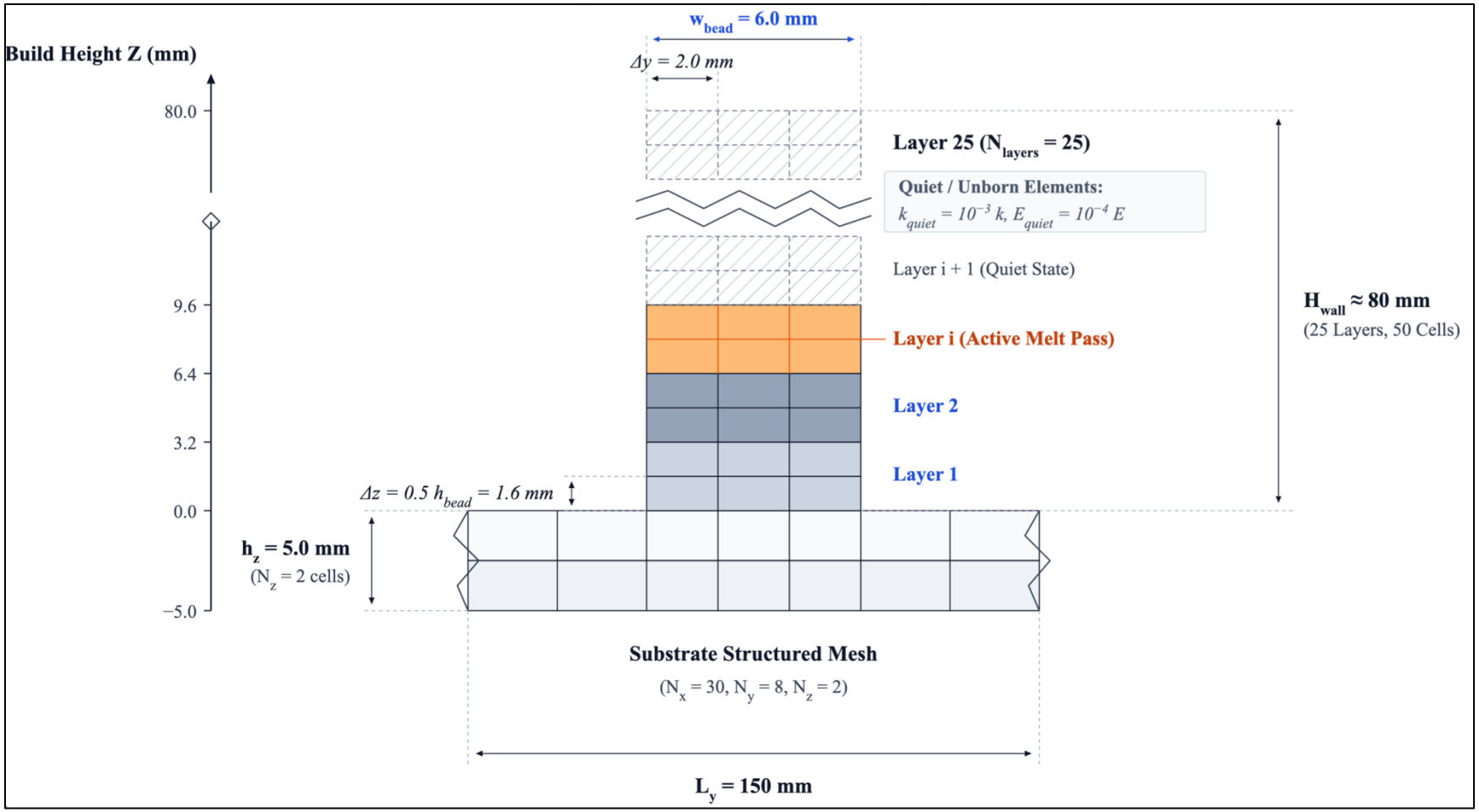


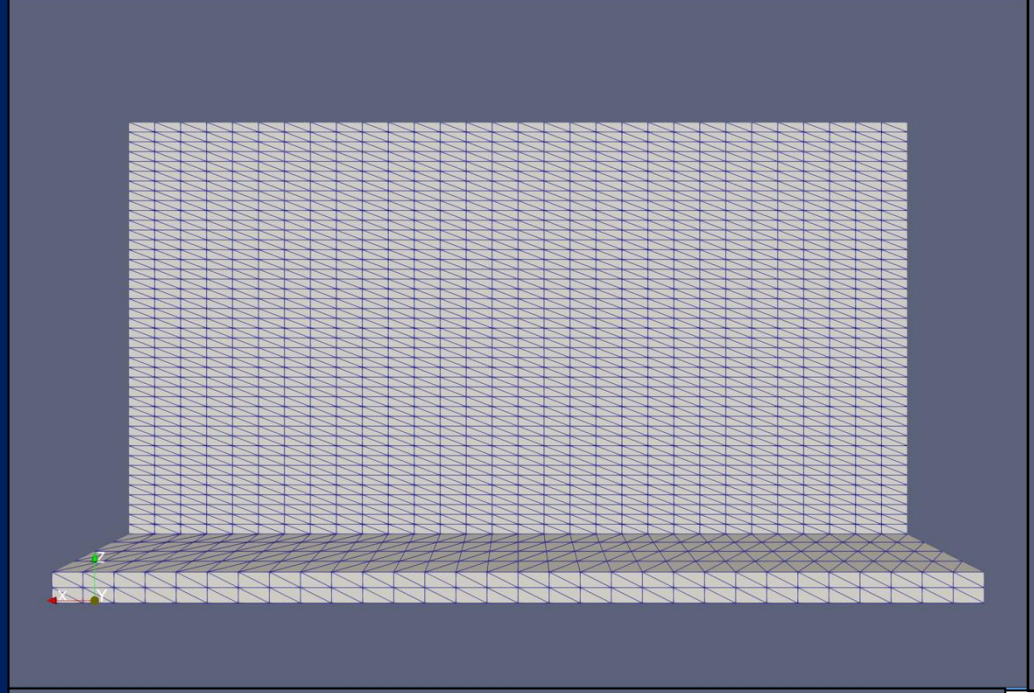
Figure : Parametric Discretization of Domain

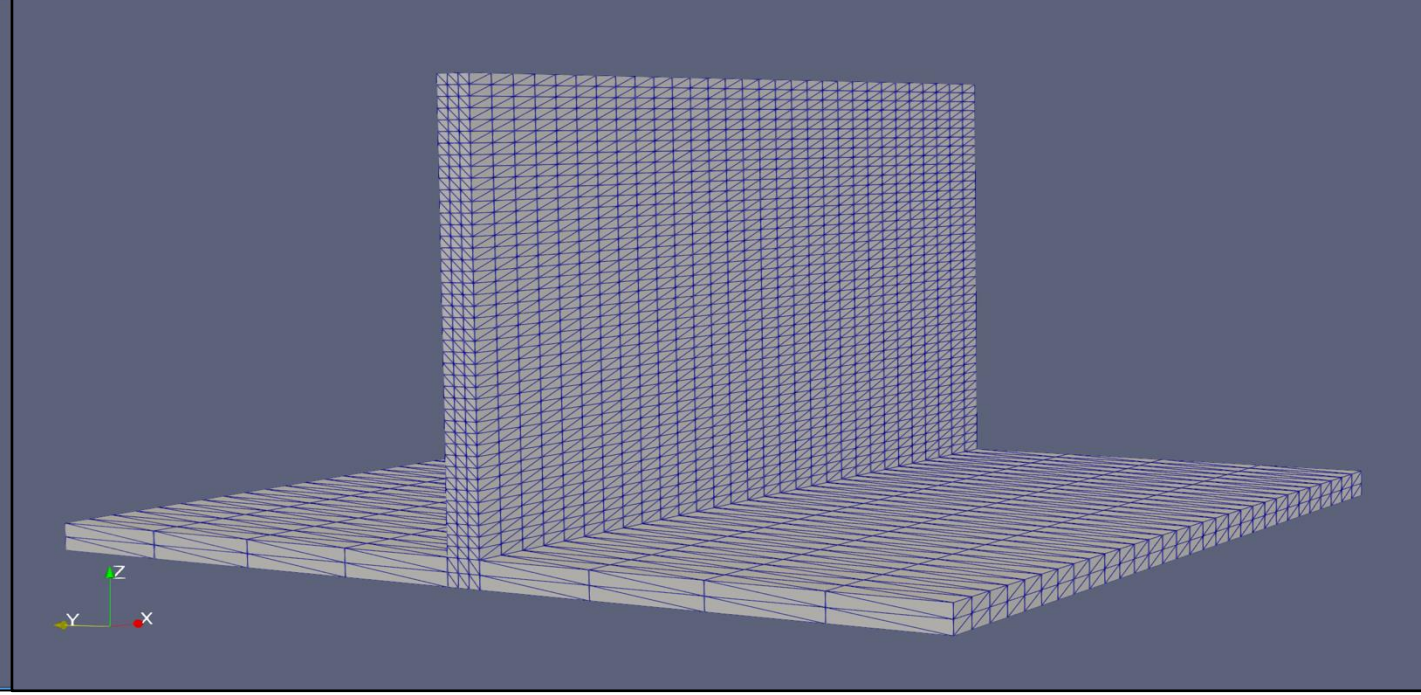
Table : Simulation Matrix with Input given to Solver

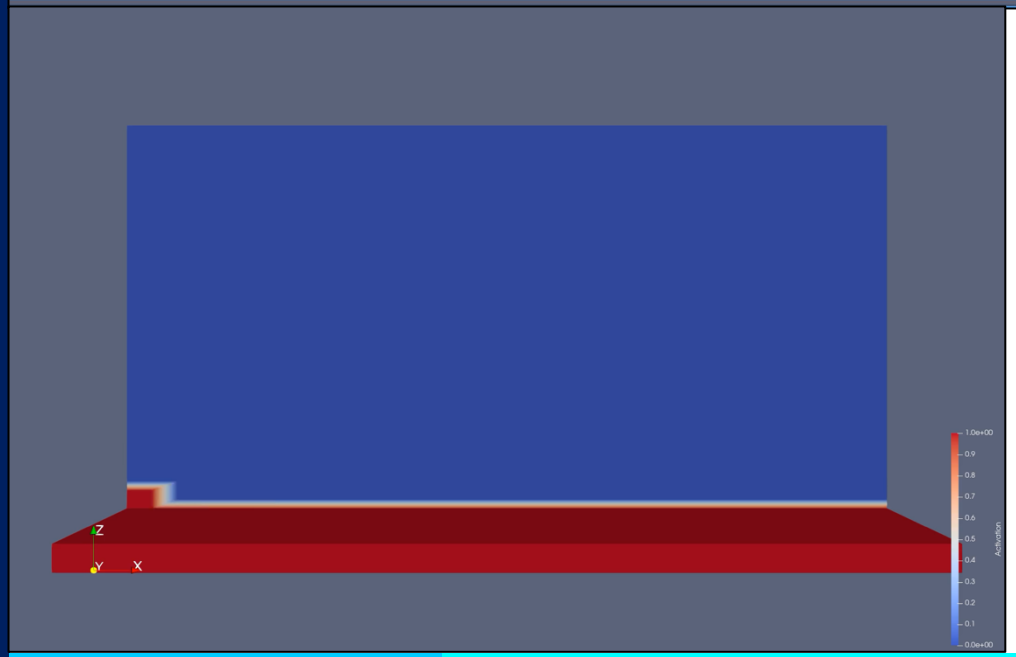
Case ID	Current [A]	Voltage [V]	Travel Speed [mm/min]	WFS [m/min]	Bead W×H [mm]	Layers [N]	Wall Ht [mm]	Net Power [W]	Linear Heat [J/mm]
1 (120A)	120	19.0	300	9.5	5.50×3.78	21	79.4	1938	387.6
2 (120A)	120	19.0	400	9.5	4.72×3.00	27	81.0	1938	290.7
3 (140A)	140	19.7	400	11.9	5.64×3.25	25	81.2	2344	351.6
4 (140A)	140	19.7	300	11.9	6.42×3.78	21	79.4	2344	468.9
5 (200A)	200	24.2	300	19.9	9.40×4.11	20	82.2	4114	822.8
6 (70A)	70	25.2	300	4.8	4.08×2.50	32	80.0	1499	299.9

Table : Mesh Information & Computational Time Taken

Case ID	Substrate Mesh	Deposit Mesh	Total Tets [elements]	Sim Time [s]	Wall Clock [min]	VTU Frames
1 (120A)	30×8×2	30×3×42	25,560	2958	20.9	882
2 (120A)	30×8×2	30×3×54	32,040	2816	26.5	910
3 (140A)	30×8×2	30×3×50	29,880	2970	22.1	846
4 (140A)	30×8×2	30×3×42	25,560	3078	20.9	883
5 (200A)	30×8×2	30×3×40	24,480	3400	19.3	846
6 (70A)	30×8×2	30×3×64	37,440	3148	45.6	1330

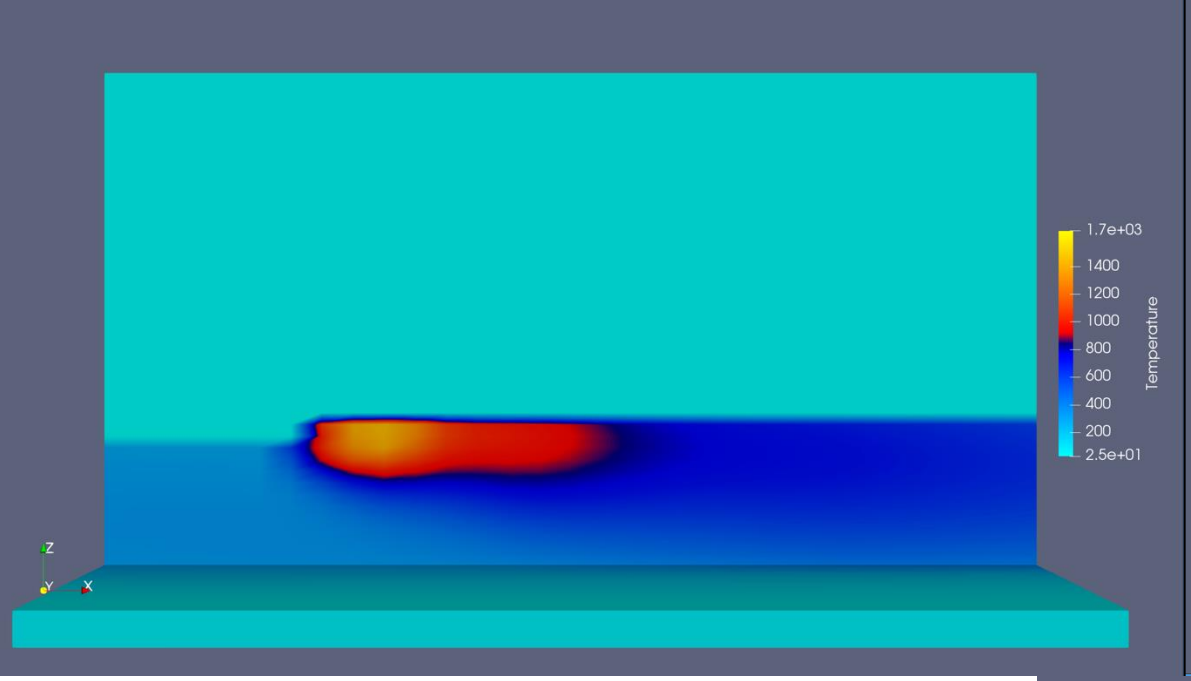


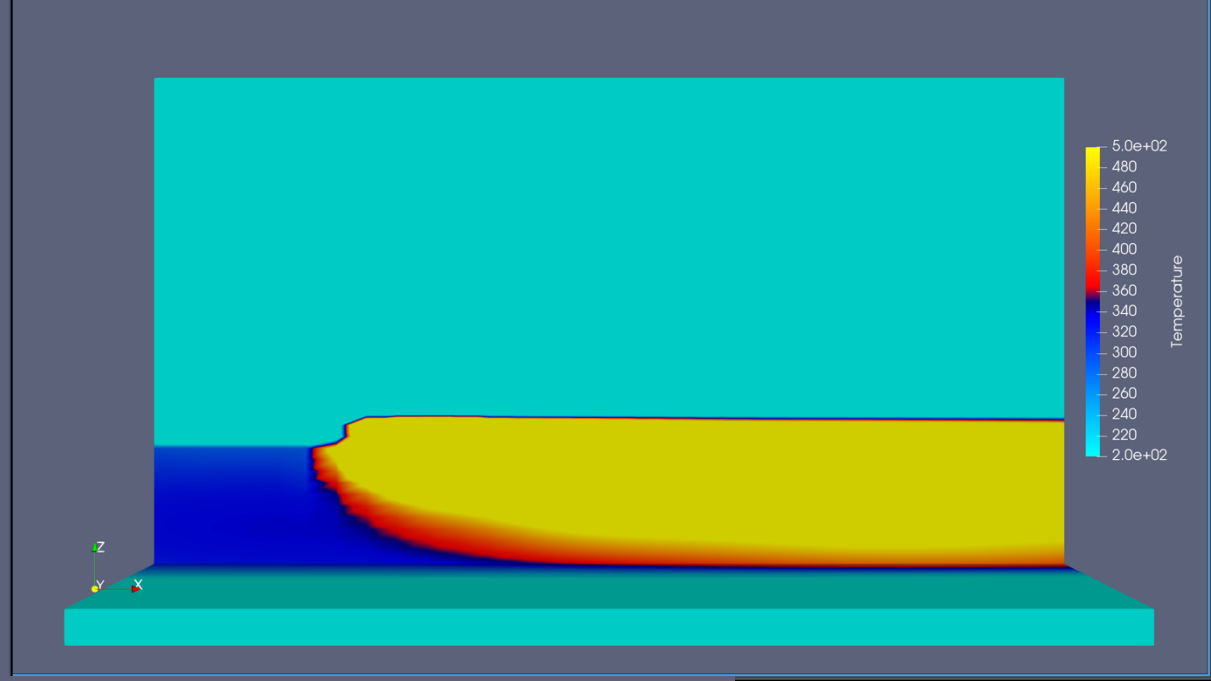




Vertical Cell Size	Δz [mm]	$\Delta z = 0.50\, h_{\text{bead}},\ \ n_{z,\text{dep}} = 2N_{\text{layers}}$
Transverse Cell Size	Δy [mm]	$\Delta y \approx 0.33\, w_{\text{bead}},\ \ n_{y,\text{dep}} = 3$
Longitudinal Cell Size	Δx [mm]	$\Delta x \approx 5.0,\ \ n_{x,\text{dep}} = 30$

Substrate (150 × 150 × 5 mm)		
Longitudinal Divisions	$n_{x,\text{base}}$ [–]	30
Transverse Divisions	$n_{y,\text{base}}$ [–]	8
Thickness Divisions	$n_{z,\text{base}}$ [–]	2





Parameter	Symbol [unit]	Value
Arc Net Power	Q_{net} [W]	$Q_{\text{net}} = \eta_{\text{arc}}\, U\, I$, $\eta_{\text{arc}} = 0.85$
Heat Source Semi-Width	b_G [mm]	$0.50\, w_{\text{bead}}$
Heat Source Penetration	c_G [mm]	$1.25\, h_{\text{bead}}$
Front Semi-Axis	a_f [mm]	$0.50\, w_{\text{bead}}$
Rear Semi-Axis	a_r [mm]	$1.25\, w_{\text{bead}}$
Front Power Fraction	f_f [–]	0.60
Rear Power Fraction	f_r [–]	1.40

Figure : Parametric Goldak Profile

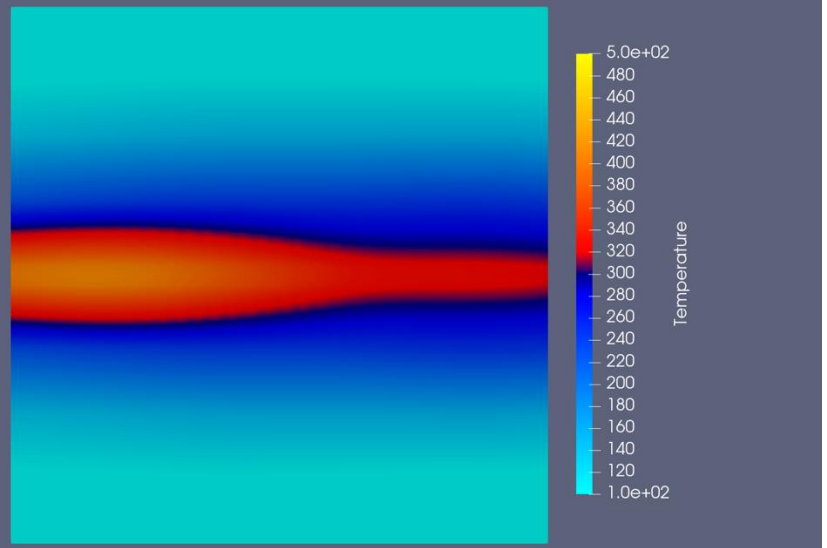
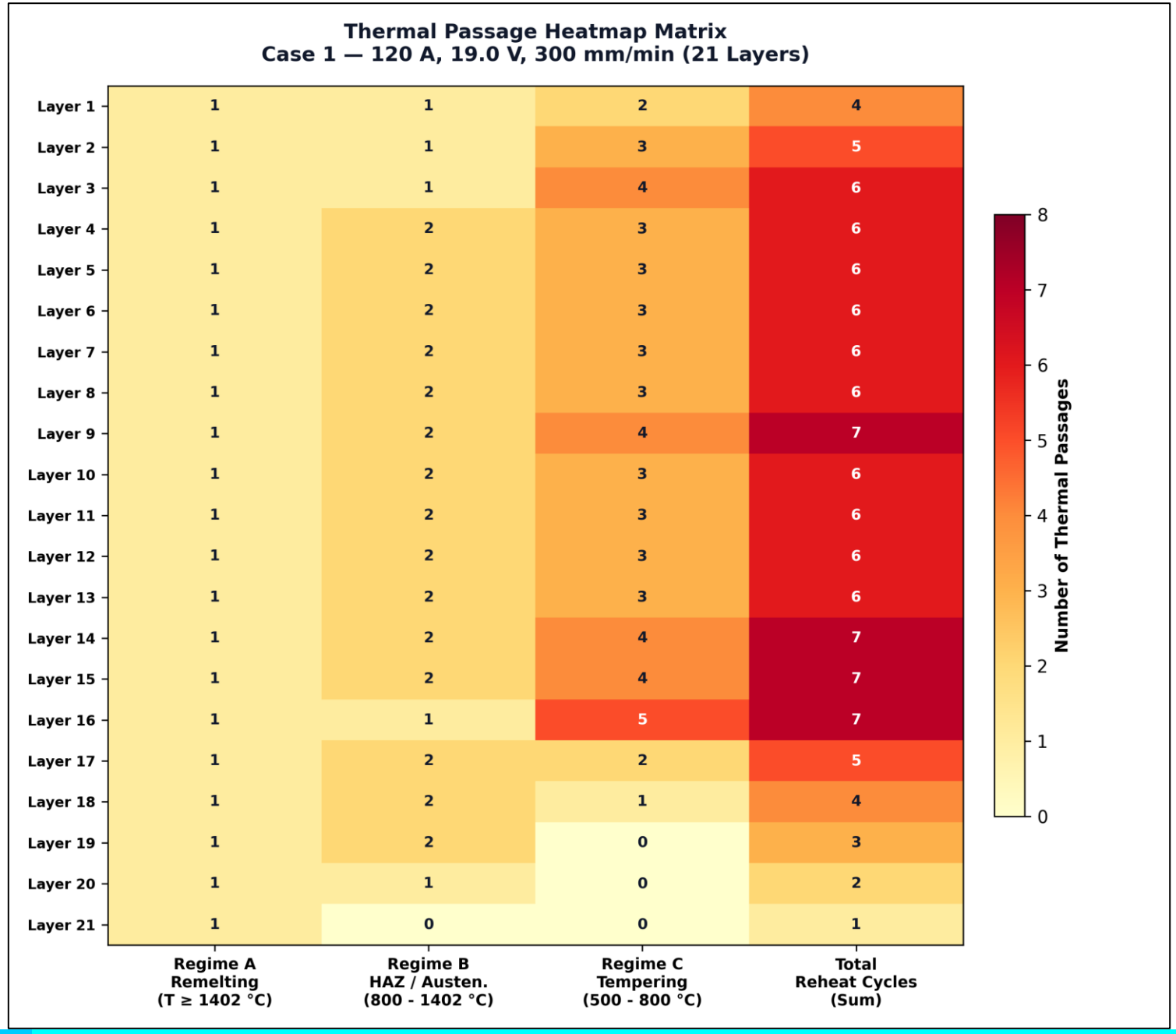
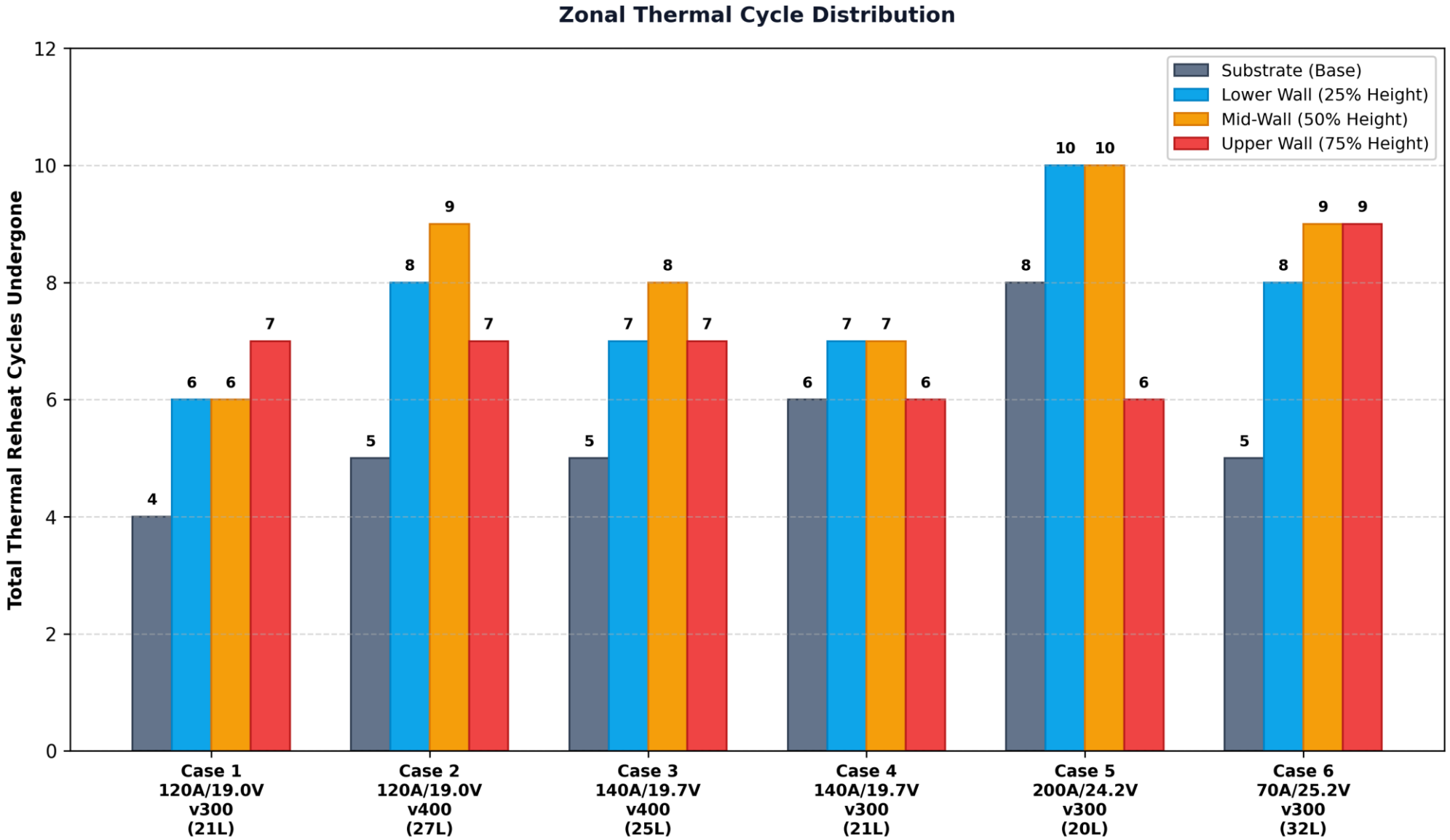


Figure : Temperature Profile at T = 273s


Thermal Passage Heatmap Matrix
Case 1 — 120 A, 19.0 V, 300 mm/min (21 Layers)
Layer 1 1 1 2 4
Layer 2 1 1 3 5
Layer 3 1 1 4 6
Layer 4 1 2 3 6
Layer 5 1 2 3 6
Layer 6 1 2 3 6
Layer 7 1 2 3 6
Layer 8 1 2 3 6
Layer 9 1 2 4 7
Layer 10 1 2 3 6
Layer 11 1 2 3 6
Layer 12 1 2 3 6
Layer 13 1 2 3 6
Layer 14 1 2 4 7
Layer 15 1 2 4 7
Layer 16 1 1 5 7
Layer 17 1 2 2 5
Layer 18 1 2 1 4
Layer 19 1 2 0 3
Layer 20 1 1 0 2
Layer 21 1 0 0 1
Regime A Remelting (T ≥ 1402 °C)
Regime B HAZ / Austen. (800 - 1402 °C)
Regime C Tempering (500 - 800 °C)
Total Reheat Cycles (Sum)
Number of Thermal Passages
0
1
2
3
4
5
6
7
8


Zonal Thermal Cycle Distribution
Total Thermal Reheat Cycles Undergone
Substrate (Base)
Lower Wall (25% Height)
Mid-Wall (50% Height)
Upper Wall (75% Height)
0
2
4
6
8
10
12
4
6
6
7
5
8
9
7
5
7
8
7
6
7
7
6
8
10
10
6
5
8
9
9
Case 1
120A/19.0V
v300
(21L)
Case 2
120A/19.0V
v400
(27L)
Case 3
140A/19.7V
v400
(25L)
Case 4
140A/19.7V
v300
(21L)
Case 5
200A/24.2V
v300
(20L)
Case 6
70A/25.2V
v300
(32L)

SUMMARY

OBJECTIVE

- Investigate WAAM of SS308L
- Investigate usage of Ultrasonic Frequency
- Investigate PFM modelling
- Investigate Macro FEM
- Multi-Physics Analysis

WORK DONE

- FEM analysis of Experiments via FreeFEM
- Developed WAAM-Framework based on FreeFEM
- Single – Bead Trials using 0.8mm and 1.2mm
- Developed Python-based Slicer and Gcode generator for Lab to eliminate manual code writing

PLANNED

- Coupling Ultrasonic / Acoustics to melt and evaluated CFF based results and melt pool evolution
- Incorporate acoustics module in WAAM Framework
- Multiwalled Prints
- Validate WAAM-Framework, incorporate stress-strain feature

THANK YOU ☺

BACKUP

Dimension	Original	Framework
Nature	One hand-written FreeFEM++ script plus a README	Layered Python application that *generates* FreeFEM++ scripts
Scale	968 lines, 2 files	28,595 lines Python + 3,571 lines `.edp` + generator (∼40×)
Workflow	Edit `.edp` → `FreeFem++ waam_multilayer.edp` → open `waam.pvd` in ParaView	GUI (12 tabs) or CLI → slice STL → mesh → generate → solve (serial/MPI) → plots + HTML report
Geometry	Hard-coded 26 × 3 mm wall on a 200 × 100 × 10 mm plate	Arbitrary STL/STEP/IGES/BREP, parametric walls, G-code-driven toolpaths
Heat source	Goldak double ellipsoid only	7 selectable models
Material	Mild steel S235 hard-coded	8-alloy JSON database, dual substrate/deposit assignment
Mechanics	Single post-hoc *elastic* solve, inherent strain	Incremental **J2 thermo-elasto-plasticity**, Newton with consistent tangent, TRIP, springback
Thermal extras	Conduction + convection	+ radiation, boiling cap, enthalpy-exact latent heat, adaptive Δt, Picard correctors, target-temperature inter-pass control
Beyond original scope	—	OpenFOAM VOF melt-pool CFD, ultrasonic (UVA) assistance, microstructure and Vickers hardness prediction
Verification	None	11-group physics suite, J2 cross-check, external CAS-DED benchmark, pytest suite
Portability	`D:\freefem++\...` hard-coded	Binary discovery + environment overrides, MPI, self-contained timestamped run directories

```
GEOMETRY → PROCESS → FEEDSTOCK → MATERIAL → TIMESTEP → G-CODE READER → MESH → ADAPTIVE dt
→ FE SPACES → THERMO-ELASTO-PLASTIC ENGINE → FORMS → PROBES → OUTPUT → RESTART → DEPOSITION
LOOP → FINAL COOLING → MECHANICAL → MICROSTRUCTURE → VALIDATION GATE → solver_results.json
```

BACKUP

1. Thermal Energy Transport and Phase Change Formulations

Non-Linear Transient Heat Conduction Equation (Strong Form)

$$\rho(T)\,c_p(T)\,\frac{\partial T}{\partial t} = \nabla\cdot\big(k(T)\,\nabla T\big) + q_{\text{source}}(\vec{x},t) - \rho(T)\,L_f\,\frac{\partial f_l}{\partial t}$$

Mushy Zone Liquid Fraction Linear Evolution

$$f_l(T) = \begin{cases} 0, & T < T_{\text{sol}} \\ \dfrac{T - T_{\text{sol}}}{T_{\text{liq}} - T_{\text{sol}}}, & T_{\text{sol}} \le T \le T_{\text{liq}} \\ 1, & T > T_{\text{liq}} \end{cases}$$

Apparent Heat Capacity Formulation

$$c_p^*(T) = c_p(T) + L_f\,\frac{\partial f_l}{\partial T} = \begin{cases} c_p(T), & T < T_{\text{sol}} \text{ or } T > T_{\text{liq}} \\ c_p(T) + \dfrac{L_f}{T_{\text{liq}} - T_{\text{sol}}}, & T_{\text{sol}} \le T \le T_{\text{liq}} \end{cases}$$

Enthalpy Formulation with Apparent Heat Capacity

$$\rho(T)\,c_p^*(T)\,\frac{\partial T}{\partial t} = \nabla\cdot\big(k(T)\,\nabla T\big) + q_{\text{source}}(\vec{x},t)$$

Finite Element Variational Weak Form (Backward Euler, $\theta = 1.0$)

$$\int_\Omega \left(\frac{\rho\,c_p^*}{\Delta t}\,T^{n+1}\,v + k(T)\,\nabla T^{n+1}\cdot\nabla v\right) d\Omega + \int_{\partial\Omega} h_{\text{eff}}\,T^{n+1}\,v\,dA = \int_\Omega \left(\frac{\rho\,c_p^*}{\Delta t}\,T^n + q_{\text{source}}^{n+1}\right) v\,d\Omega + \int_{\partial\Omega} h_{\text{eff}}\,T_\infty\,v\,dA$$

Linearized Stefan-Boltzmann Radiation Coefficient

$$h_{\text{rad}}(T) = \varepsilon\,\sigma_{\text{SB}}\,(T + T_\infty)\,(T^2 + T_\infty^2) \approx 4\,\varepsilon\,\sigma_{\text{SB}}\,T_{\text{ref}}^3$$

Effective Combined Surface Convective-Radiative Heat Transfer Coefficient

$$h_{\text{eff}} = h_{\text{conv}} + h_{\text{rad}}$$

Boiling Temperature Surface Evaporation Cap

$$T(\vec{x},t) = \min\big(T(\vec{x},t),\,T_{\text{boil}}\big) \qquad (T_{\text{boil}} = 2862\,^\circ\text{C})$$

Latent Heat of Vaporization Dissipation Accounting

$$E_{\text{cap}} = \int_\Omega \rho\,c_p^*\,\max(T - T_{\text{boil}},\,0)\,d\Omega$$

BACKUP

2. Moving Volumetric Heat Source Formulations (Goldak Double-Ellipsoid)

Lagrangian-to-Eulerian Torch Coordinate Transformation

$$\xi = (x - x_0) - v_{\text{scan}}\,(t - t_{\text{start}})$$

Front Semi-Ellipsoid Heat Flux Density Distribution ($\xi \geq 0$)

$$q_f(\xi, y, z) = \frac{6\sqrt{3}\, f_f\, Q_{\text{net}}}{\pi\sqrt{\pi}\, a_f\, b_G\, c_G} \exp\left(-3\,\frac{\xi^2}{a_f^2} - 3\,\frac{y^2}{b_G^2} - 3\,\frac{(z - z_0)^2}{c_G^2}\right)$$

Rear Semi-Ellipsoid Heat Flux Density Distribution ($\xi < 0$)

$$q_r(\xi, y, z) = \frac{6\sqrt{3}\, f_r\, Q_{\text{net}}}{\pi\sqrt{\pi}\, a_r\, b_G\, c_G} \exp\left(-3\,\frac{\xi^2}{a_r^2} - 3\,\frac{y^2}{b_G^2} - 3\,\frac{(z - z_0)^2}{c_G^2}\right)$$

Heat Flux Spatial Continuity Constraint at Arc Center Plane ($\xi = 0$)

$$q_f(0, y, z) = q_r(0, y, z) \iff \frac{f_f}{a_f} = \frac{f_r}{a_r} \implies f_r = f_f\left(\frac{a_r}{a_f}\right)$$

Total Power Energy Conservation Constraint

$$\int_{-\infty}^{0}\iint_{-\infty}^{\infty} q_r\, dy\, dz\, d\xi + \int_{0}^{\infty}\iint_{-\infty}^{\infty} q_f\, dy\, dz\, d\xi = Q_{\text{net}} \iff f_f + f_r = 2.0$$

Energy Partition Fractions Solution

$$f_f = \frac{2\,a_f}{a_f + a_r}, \qquad f_r = \frac{2\,a_r}{a_f + a_r}$$

Net Absorbed Arc Thermal Power

$$Q_{\text{net}} = \eta_{\text{arc}}\, U\, I$$

Transverse Semi-Axis Sizing from Bead Width

$$b_G = 0.50\, w_{\text{bead}}$$

Penetration Semi-Axis Sizing from Bead Height

$$c_G = 1.25\, h_{\text{bead}}$$

Longitudinal Front Semi-Axis Sizing

$$a_f = 0.50\, w_{\text{bead}}$$

Longitudinal Rear Semi-Axis Sizing

$$a_r = 1.25\, w_{\text{bead}} = 2.50\, a_f$$

Dimensionless Energy Fractions for 2.5x Rear Elongation

$$f_f = \frac{2(0.50\, w_{\text{bead}})}{0.50\, w_{\text{bead}} + 1.25\, w_{\text{bead}}} = \frac{1.00}{1.75} \approx 0.60, \qquad f_r = \frac{2(1.25\, w_{\text{bead}})}{0.50\, w_{\text{bead}} + 1.25\, w_{\text{bead}}} = \frac{2.50}{1.75} \approx 1.40$$

Peak Volumetric Power Density at Torch Axis ($\xi = 0, y = 0, z = z_0$)

$$q_0 = \frac{6\sqrt{3}\, f_f\, Q_{\text{net}}}{\pi\sqrt{\pi}\, a_f\, b_G\, c_G} = \frac{6\sqrt{3}\, f_r\, Q_{\text{net}}}{\pi\sqrt{\pi}\, a_r\, b_G\, c_G}$$

BACKUP

3. Quiet Element Activation and Inactive State Suppression

Element Activation State Heaviside Function

$$\psi(\vec{x},t) = \begin{cases} 1, & \vec{x} \in \Omega_{\text{activated}}(t) \\ 0, & \vec{x} \in \Omega_{\text{quiet}}(t) \end{cases}$$

Progressive Kinematic Torch Activation Condition

$$\psi_e(t) = 1 \quad \Longleftrightarrow \quad x_{\text{torch}}(t) \geq x_e \quad \text{and} \quad z_{\text{torch}}(t) \geq z_e$$

Quiet Element Thermal Conductivity Degradation

$$k_{\text{eff}}(\vec{x},T) = \psi(\vec{x},t)\,k(T) + \big(1-\psi(\vec{x},t)\big)\cdot 10^{-4}\,k(T)$$

Volumetric Heat Capacity Conservation in Quiet Elements

$$(\rho\, c_p^*)_{\text{eff}}(\vec{x},T) = \rho(T)\, c_p^*(T)$$

Quiet Element Mechanical Stiffness Degradation

$$E_{\text{eff}}(\vec{x},T) = \psi(\vec{x},t)\,E(T) + \big(1-\psi(\vec{x},t)\big)\cdot 10^{-4}\,E(T)$$

Quiet Element Yield Stress Suppression

$$\sigma_{y,\text{eff}}(\vec{x},T) = \psi(\vec{x},t)\,\sigma_y(T) + \big(1-\psi(\vec{x},t)\big)\cdot 10^{-3}\,\sigma_y(T)$$

BACKUP

4. Thermal Boundary Energy Balance Equations

Top Active Surface Energy Balance (Influx, Convection, and Radiation)

$$-k(T)\,\nabla T\cdot\vec{n} = q_{\text{source}}(\xi,y,z) - h_{\text{conv}}\,(T-T_{\infty}) - \varepsilon\,\sigma_{\text{SB}}\,(T^4-T_{\infty}^4)$$

Lateral Wall Faces Convective and Radiative Dissipation (Robin BC)

$$-k(T)\,\frac{\partial T}{\partial n} = h_{\text{conv}}\,(T-T_{\infty}) + \varepsilon\,\sigma_{\text{SB}}\,(T^4-T_{\infty}^4)$$

Multi-Bead Overlapping Lateral Solid Conduction Continuity

$$-k_1\,\nabla T_1\cdot\vec{n}_1 = -k_2\,\nabla T_2\cdot\vec{n}_2, \qquad T_1 = T_2$$

Lateral Chill Block / Copper Shoe Contact Heat Flux

$$-k(T)\,\nabla T\cdot\vec{n} = h_{\text{chill}}\,(T-T_{\text{chill}})$$

Substrate Bottom Contact Heat Conduction with Fixture Table

$$-k(T)\,\nabla T\cdot\vec{n} = h_{\text{contact}}\,(T-T_{\text{table}})$$

Initial Domain Temperature Field

$$T(\vec{x},0) = T_{\text{amb}} = 25\,^{\circ}\text{C}$$

BACKUP

5. Feedstock Kinematics, Mass Conservation, and Bead Dimensions

Volumetric Wire Feed Delivery Rate

$$Q_{\text{wire}} = \left(\frac{\pi d_{\text{wire}}^2}{4} \right) v_{\text{wire}}$$

Theoretical Bead Cross-Sectional Area (Volume Conservation)

$$A_{\text{bead}} = \frac{Q_{\text{wire}}}{v_{\text{scan}}} = \frac{\pi d_{\text{wire}}^2 v_{\text{wire}}}{4 v_{\text{scan}}}$$

Parabolic Bead Profile Area Relation

$$A_{\text{bead}} = \frac{2}{3} w_{\text{bead}} h_{\text{bead}}$$

Parabolic Bead Reinforcement Height

$$h_{\text{bead}} = \frac{3 A_{\text{bead}}}{2 w_{\text{bead}}} = \frac{3 \pi d_{\text{wire}}^2 v_{\text{wire}}}{8 w_{\text{bead}} v_{\text{scan}}}$$

Semi-Elliptical Bead Reinforcement Height

$$h_{\text{bead}} = \frac{4 A_{\text{bead}}}{\pi w_{\text{bead}}} = \frac{d_{\text{wire}}^2 v_{\text{wire}}}{w_{\text{bead}} v_{\text{scan}}}$$

Subsequent Pass Layer Step Height (Remelting Overlap)

$$h_{\text{subseq}} = \eta_{\text{remelt}} h_{\text{bead}} \qquad (\eta_{\text{remelt}} \approx 0.80)$$

Total Multi-Layer Wall Height Accumulation

$$H_{\text{total}} = h_{\text{bead_first}} + (N_{\text{layers}} - 1) h_{\text{subseq}}$$

BACKUP

6. Linear Heat Input and Analytic Cooling Times

Net Linear Heat Input per Unit Travel

$$H_{\text{net}} = \frac{Q_{\text{net}}}{v_{\text{scan}}} = \frac{\eta_{\text{arc}}\, U\, I}{v_{\text{scan}}}$$

2D Thin-Plate Cooling Time ($\Delta t_{8/5}^{(2D)}$)

$$\Delta t_{8/5}^{(2D)} = \frac{(H_{\text{net}})^2}{4\,\pi\, k\, \rho\, c_p\, w_{\text{bead}}^2} \left[\left(\frac{1}{500 - T_0} \right)^2 - \left(\frac{1}{800 - T_0} \right)^2 \right]$$

3D Thick-Plate Cooling Time ($\Delta t_{8/5}^{(3D)}$)

$$\Delta t_{8/5}^{(3D)} = \frac{H_{\text{net}}}{2\,\pi\, k} \left[\frac{1}{500 - T_0} - \frac{1}{800 - T_0} \right]$$

Relative Plate Thickness Dimensional Transition Parameter

$$\theta_t = w_{\text{bead}} \sqrt{\frac{\rho\, c_p\, (500 - T_0)}{H_{\text{net}}}}$$

BACKUP

7. Solidification Dynamics and Microstructure Metrics

Thermal Gradient Vector and Magnitude at Solidus Front

$$\vec{G} = \nabla T, \qquad G = \|\nabla T\| = \sqrt{\left(\frac{\partial T}{\partial x}\right)^2 + \left(\frac{\partial T}{\partial y}\right)^2 + \left(\frac{\partial T}{\partial z}\right)^2}$$

Solidification Front Normal Growth Velocity

$$R = v_{\text{scan}} \cos\theta = v_{\text{scan}} \left(\frac{-\partial T/\partial x}{\|\nabla T\|}\right)$$

Solidification Cooling Rate

$$\dot{T} = \left|\frac{\partial T}{\partial t}\right| = G \cdot R = -v_{\text{scan}} \frac{\partial T}{\partial x}$$

Solidification Morphology Ratio

$$\frac{G}{R} = \frac{\|\nabla T\|^2}{-v_{\text{scan}} \left(\partial T/\partial x\right)}$$

Secondary Dendrite Arm Spacing (SDAS, λ_2)

$$\lambda_2 = K \left(\dot{T}\right)^{-n}$$

Columnar-to-Equiaxed Transition (Hunt CET Criterion)

$$\Phi_{\text{eq}} = \exp\left(-\frac{4\pi}{3} N_0 \left[\frac{G}{\Delta T_{\text{crit}}}\right]^{-3}\right)$$

BACKUP

8. Quasi-Static Mechanics and Continuum Kinematics

Quasi-Static Momentum Balance Equation (Strong Form)

$$\nabla \cdot \sigma + \vec{b} = \vec{0} \qquad (\vec{b} \approx \vec{0})$$

Weak Variational Form of Momentum Balance

$$\int_{\Omega} \sigma : \nabla \vec{w}\, d\Omega = \int_{\partial\Omega} \vec{t} \cdot \vec{w}\, dA + \int_{\Omega} \vec{b} \cdot \vec{w}\, d\Omega \quad \forall \vec{w} \in \mathscr{V}_0$$

Infinitesimal Strain-Displacement Relation

$$\varepsilon = \frac{1}{2}\left(\nabla \vec{u} + (\nabla \vec{u})^T\right), \qquad \varepsilon_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)$$

Additive Strain Tensor Decomposition

$$\varepsilon = \varepsilon^e + \varepsilon^p + \varepsilon^{\text{th}}$$

Isotropic Thermal Expansion Strain Tensor

$$\varepsilon^{\text{th}} = \alpha_{\text{sec}}(T)\,(T - T_{\text{ref}})\,\mathbf{I}$$

Secant Thermal Expansion Coefficient Definition

$$\alpha_{\text{sec}}(T) = \frac{1}{T - T_{\text{ref}}} \int_{T_{\text{ref}}}^{T} \alpha_{\text{inst}}(T')\, dT'$$

BACKUP

9. Constitutive Elastoplasticity and Radial Return Mapping (J_2 Flow)

Generalized Hooke's Law for Isotropic Elasticity

$$\sigma = \mathbf{C}(T) : \varepsilon^e = 2\,G(T)\,\mathbf{e}^e + K(T)\,\mathrm{tr}(\varepsilon^e)\,\mathbf{I}$$

Elastic Shear and Bulk Moduli

$$G(T) = \frac{E(T)}{2(1+\nu)}, \qquad K(T) = \frac{E(T)}{3(1-2\nu)}$$

Deviatoric Cauchy Stress Tensor

$$\mathbf{s} = \sigma - \frac{1}{3}\,\mathrm{tr}(\sigma)\,\mathbf{I}$$

Second Invariant of Deviatoric Stress (J_2)

$$J_2 = \frac{1}{2}\,\mathbf{s} : \mathbf{s} = \frac{1}{2}\,s_{ij}\,s_{ij}$$

Von Mises Equivalent Stress

$$\sigma_{\mathrm{eq}} = \sqrt{3J_2} = \sqrt{\frac{3}{2}\,\mathbf{s} : \mathbf{s}} = \sqrt{\frac{3}{2}\,s_{ij}\,s_{ij}}$$

Von Mises Yield Function with Linear Isotropic Hardening

$$f(\sigma, p) = \sigma_{\mathrm{eq}} - \sigma_y(T, p) \leq 0$$

Current Yield Strength Evolution

$$\sigma_y(T, p) = \sigma_{y0}(T) + H'(T)\,p$$

Prandtl-Reuss Associative Plastic Flow Rule

$$\dot{\varepsilon}^p = \dot{\gamma}\,\frac{\partial f}{\partial \sigma} = \dot{\gamma}\,\sqrt{\frac{3}{2}}\,\frac{\mathbf{s}}{\|\mathbf{s}\|} = \frac{3}{2}\,\dot{p}\,\frac{\mathbf{s}}{\sigma_{\mathrm{eq}}}$$

Accumulated Equivalent Plastic Strain Rate

$$\dot{p} = \sqrt{\frac{2}{3}\,\dot{\varepsilon}^p : \dot{\varepsilon}^p}$$

BACKUP

10. Mechanical Boundary Constraints

Rigid Baseplate Clamping Constraint (Dirichlet BC)

$$\vec{u}(\vec{x}) = \vec{0} \iff u_x = 0, \quad u_y = 0, \quad u_z = 0 \qquad \text{on } \Gamma_{\text{clamp}}$$

Isostatic (3-2-1) Minimal Kinematic Support Constraint

$$\vec{u}(0,0,0) = \vec{0}, \qquad u_y(L_x,0,0) = u_z(L_x,0,0) = 0, \qquad u_z(0,L_y,0) = 0$$

Unilateral Baseplate Contact Support (Signorini Condition)

$$u_z \geq 0, \qquad \sigma_{zz} \geq 0, \qquad u_z \cdot \sigma_{zz} = 0 \qquad \text{at } z = 0$$

Free Surface Traction-Free Condition (Neumann BC)

$$\sigma \cdot \vec{n} = \vec{0} \qquad \text{on } \Gamma_{\text{free}}$$

11. Finite Element Discretization and Numerical Stability Criteria

Vertical Deposit Element Sizing (Δz_{dep})

$$\Delta z_{\text{dep}} = 0.50\, h_{\text{bead}} \implies n_{z,\text{dep}} = 2 \times N_{\text{layers}}$$

Transverse Deposit Element Sizing (Δy_{dep})

$$\Delta y_{\text{dep}} \approx 0.33\, w_{\text{bead}} \implies n_{y,\text{dep}} = 3$$

Longitudinal Deposit Element Sizing (Δx_{dep})

$$\Delta x_{\text{dep}} = \frac{L_x}{n_{x,\text{dep}}} = \frac{150\,\text{mm}}{30} = 5.0\,\text{mm}$$

Thermal Courant-Friedrichs-Lewy (CFL) Advection Alignment

$$\text{CFL}_{\text{thermal}} = \frac{v_{\text{scan}}\, \Delta t_{\text{weld}}}{\Delta x_{\text{dep}}} \approx \frac{(5.0\,\text{mm/s})\,(1.0\,\text{s})}{5.0\,\text{mm}} = 1.0$$